

What Works for Working Couples? Schedule Regularity and Gender Equality*

Ludovica Ciasullo

Martina Uccioli

July 23, 2026

Abstract

We provide the first causal evidence that changes to work arrangements – in the form of greater schedule regularity – can reduce the gender earnings gap. The Australian 2009 Fair Work Act explicitly entitled parents of young children to request a change in work arrangements. Leveraging variation in the timing of the law, timing of childbirth, and the bite of the law across different occupations and industries, we establish three main results. First, new mothers used the Fair Work Act to maintain a regular schedule while reducing hours upon childbirth. Second, thanks to increased regularity, new mothers' hours worked increased by 53%. Third, while this increase in maternal labor supply resulted in a significant shift towards equality in the female- and male-shares of household income, we do not observe any changes in the female share of home production.

Keywords: Labor Supply, Work Arrangements, Home Production

JEL: J16, J22, J18, J81

*Ciasullo: Cornerstone Research, ludovica.ciasullo@gmail.com ; Uccioli: University of Nottingham, CESifo, RFBerlin, and IZA, martina.uccioli@nottingham.ac.uk. First version: October 20, 2022. Older versions of this paper were previously circulated under the title “What Works for Working Mothers? A Regular Schedule Lowers the Child Penalty” and “What Works for Working Couples? Work Arrangements, Maternal Labor Supply, and the Division of Home Production”. We are grateful to Daron Acemoglu, Amy Finkelstein, Simon Jäger, Petra Moser, Debraj Ray, and Martin Rotemberg for guidance and advice throughout this project. We are especially grateful to Mark Wooden for discussions on the institutional context that pointed us in the right direction. We thank Josh Angrist, David Autor, Maria Balgova, Mattia Bertazzini, Vincenzo Carrieri, Alden Cheng, Jonathan Cohen, Viola Corradini, Angela Crema, Marc de la Barrera i Bardalet, Enrico de Magistris, Isabel Di Tella, Raquel Fernández, Joel Flynn, Michael Gilraine, Valentina Goetz, Claudia Goldin, Jonathan Gruber, Ahmet Gulek, Clemence Idoux, Hannah Illing, Ingo Isphording, Jonas Jessen, Jeanne Lafortune, Chitra Marti, Matías Morales, Claudia Olivetti, Abby Ostriker, Helena Pedrotti, James Poterba, Charlie Rafkin, Ulf Rinne, Ursina Schaeede, Frank Schilbach, Garima Sharma, Jason Sockin, Anna Stansbury, Teresa Steininger, Diana Sverdlin Lisker, Marco Tabellini, Nagisa Tadjfar, Ishaana Talesara, Sharon Traiberman, Sean Wang, Miriam Wüst, Samuel Young, Whitney Zhang, all the participants at the 2025 NBER Summer Institute, 2025 Barcelona Summer Forum, 2025 CESifo Area Conference on Labor Economics, 2024 Badolato Labour and Organizations Workshop, 2023 HILDA Survey Research Conference, 2023 Stanford SITE Gender, 2023 Meeting of SEHO, NYU Applied Micro Lunch, MIT Public Finance and Labor Lunch, and the seminar participants at Harvard Business School, Stanford University, the Center for Economic Performance at the London School of Economics, Tufts University, Carlson School of Management, Collegio Carlo Alberto, University of Naples Federico II, Catholic University of Milan, St. Andrews University, University of Nottingham, Bank of Italy, IESE business school, Carnegie Mellon University's Heinz College, Bank of Spain, Federal Reserve Board of Governors, University of York, WU Vienna University of Economics and Business, Bonn BGSE Applied Microeconomics Workshop, ECARES at the Université libre de Bruxelles, KOF Institute at ETH Zurich, RWI Berlin, Loughborough University, and the Office of Tax Analysis at the Treasury for helpful comments. Uccioli is grateful to Jerry Hausman and the NBER Retirement and Disability Research Center for financial support. Disclaimers are at the end of the draft.

1 Introduction

The notion that control over one’s time at work is crucial for gender equality underpins many public debates: policymakers on both sides of the Atlantic advocate to give workers more control over their schedules in order to help mothers’ careers.¹ Indeed, over the past decades, occupations where hours have become more predictable have seen a reduction in the gender wage gap (Goldin and Katz, 2016). This could be causal: unpredictable hours might prevent mothers from securing reliable childcare and thus constrain their labor supply and career prospects. However, these patterns might also reflect sorting, or reverse causality: we lack causal evidence that schedule regularity and predictability – an amenity 60% of the American workforce does not enjoy (Ganong, Noel, Patterson, Vavra, and Weinberg, 2025) – affect the gender earnings gap.

We are the first to demonstrate a causal link between schedule regularity and the gender earnings gap. In this paper, we show that, when given the possibility of keeping a regular week-to-week schedule upon reducing work hours after childbirth, new mothers increase their hours worked by 53%; this translates into a 19% lower earnings gap relative to fathers. Given that the earnings gap between mothers and fathers accounts for two thirds of the overall gender earnings gap in our setting, granting mothers a regular schedule can be a powerful tool for reducing the overall earnings inequality.

We identify this causal link thanks to the variation provided by the Australian 2009 Fair Work Act and the richness of the available Australian data. The 2009 Fair Work Act entitled parents to request a change in work arrangements, thus generating quasi-exogenous variation in the work arrangements available to mothers – most notably, schedule regularity – that we will exploit in our analysis. The *Household, Income, and Labour Dynamics in Australia* (HILDA) dataset provides very rich longitudinal data on work arrangements, time use, and family linkages for over 20 years: this allows us to directly observe the work environment and time allocation of both partners in the couple, and how they change with legislative changes.

We first show that the policy changed schedule regularity for mothers. As in several

¹For example, the US Congress is debating the “Schedules That Work Act” (House bill H.R.6786) introduced to curb scheduling practices that make it “impossible for workers to arrange child care” and to help “people meet their responsibilities at work and at home” (Warren-DeLauro press release, 12/23/25). In Europe, since August 2022 all European Union member states must apply the “Directive on work-life balance for parents and carers” (Directive (EU) 2019/1158) which establishes rights “such as the right to request flexible working arrangements, which will help people develop their careers and family life without having to sacrifice either” (European Commission press release, 8/2/22).

other countries, Australian women reduce hours of work when they have children, but, before the law, they would do so by moving to a casual contract. Casual contracts pay more per hour, but are extremely irregular, with no commitment on either the employer's or the employee's side. The Fair Work Act strengthened mothers' ability to reduce hours of work within their permanent contracts, which are associated with a regular schedule, instead of moving to a casual contract and an irregular schedule. The take-up was large: the probability of new mothers having a regular schedule increased by 45% after the Fair Work Act.

While casual contracts are a peculiarity of the Australian system, irregular and unpredictable schedules are not. For example, 30% of all vacancies in the UK in 2019 came without a fixed schedule (Adams, Balgova, Qian, and Waters, Forthcoming); in the US, 60% of the workforce is paid hourly and experiences large fluctuations in hours (Ganong et al., 2025). Thus, the lessons we learn from this analysis have a significance that goes beyond the specific setting: they suggest that, for example, the predictive scheduling laws many US localities have been introducing,² although not directly targeting it, might prove effective at reducing the gender earnings gap.

Second, we present our main results, which we obtain through two complementary empirical strategies. In the first, we leverage variation in the timing of first childbirth to show the aggregate changes to maternal labor supply the Fair Work Act caused. In the second, we exploit additional variation from differential occupational exposure to the law to argue that the changes in mothers' schedule regularity do directly cause the observed changes in labor supply, and to zoom into mechanisms and margins of adjustment.

We show that having access to a regular and predictable schedule when reducing hours upon childbirth dramatically increases labor supply for treated mothers. After the law, women still move from full time to part time upon motherhood, but, with the newly available schedule regularity, they work 53% more hours than their pre-reform counterparts. This increase in hours is *not* associated with a change in the gap between desired and actual hours suggesting that, when a regular schedule is available, new mothers want and do work more hours (rather than, for example, facing a different labor demand or being closer to a pre-existing ideal number of hours).

Giving mothers schedule regularity in their less-than-full-time contracts does not have negative consequences on mothers' careers; if anything, estimates suggest a medium-term improvement. First, although the law made less-than-full-time arrangements more attractive, we don't find a reduction in the probability of working full time: the counterfactual

²<https://www.hrdive.com/news/a-running-list-of-states-and-localities-with-predictive-scheduling-mandates/540835/>

for these mothers now in less-than-full-time permanent and regular jobs would have been a less-than-full-time casual and irregular job (or no job altogether), *not* full time employment. Second, we estimate, albeit with noise, that the longer hours worked by the most exposed mothers make them more likely to hold supervisory roles in the medium run.

Finally, we turn to the household. The documented increase in labor supply led to a large increase in maternal income. Did it then reflect back to task allocation within the household, as standard economic models would predict? While treated mothers' contribution to household income increases dramatically, we find no significant changes in the female share of home production. In other words, we find that a sizable increase in the female share of income does not lead to any decrease in the disproportionate share of home production women shoulder. This suggests that relative income is not the only determinant of how tasks are split within the couple, and that something else, possibly gender norms, plays an important role.

Related Literature Our work contributes to the literature linking occupational characteristics and the gender earnings gap. At the aggregate level, occupations where workers have greater control over their work time exhibit smaller gender wage gaps (Goldin, 2014); in a field experiment, mothers of young children exhibit a disproportionately high willingness-to-pay to avoid employer's discretion over their schedule (Mas and Pallais, 2017). While two papers we are aware of apply this notion to maternal labor supply, documenting smaller child penalties in occupations that give workers more control over their time (Bang, 2021; Bütikofer, Jensen, and Salvanes, 2018), we are the first to establish a causal link between schedule regularity and the gender earnings gap.

We also contribute to the discussion around household division of tasks and whether it responds to economic incentives. Several studies found that women carry out a greater share of household tasks at every percentile of the within-couple relative income distribution (Bertrand, Kamenica, and Pan, 2015; Hancock, Lafortune, and Low, 2025; Jessen, Schweighofer-Kodritsch, Weinhardt, and Berkes, 2024; Siminski and Yetsenga, 2022). However, this distribution is likely to be endogenous to choices made before marriage and childbirth (like education and job sorting). Instead, we rely on a quasi-exogenous change in the relative earnings distribution within the family to show that, indeed, bringing home a greater share of household income does not lead mothers to shoulder a smaller share of home production.³

³See Ichino, Olsson, Petrongolo, and Skogman Thoursie (2026) for related evidence in the context of Sweden (on a smaller scale).

The remainder of the paper is organized as follows. Section 2 describes the data and the sample. Section 3 shows that the Fair Work Act increased schedule regularity for mothers, both qualitatively and quantitatively. Section 4 quantifies the impact of this change in schedule regularity on maternal labor supply and other work-related outcomes of mothers. Section 5 expands the view to the household and shows that the reform-induced increase in female share of household income did not translate into a change in the female share of home production. Section 6 concludes.

2 Data and Sample Selection

The Household, Income, and Labour Dynamics in Australia (HILDA) Survey (Department of Social Services and Melbourne Institute of Applied Economic and Social Research, 2020) contains a rich set of information on a representative sample of Australian households since 2001. For each of the approximately 8,000 sampled households,⁴ everyone above 15 years old is interviewed, but basic demographics are also collected for younger members of the household. Individuals within families are followed over time, and family linkages are codified, which allows us to match each individual with their spouse or *de facto* partner and to identify when children are born. The design of the HILDA Survey is detailed in Watson and Wooden (2012). We limit the sample to the years 2001-2019 to avoid confounders from the COVID pandemic.

Variables Definition Our analysis relies on measures of work arrangements and of time use. Key variables in our analysis are the type of contract (whether casual or permanent), whether a schedule is regular (working from Monday to Friday or four weekdays on a regular daytime schedule) or irregular (being on call), whether the job is full-time (35 hours per week or more), less-than-full-time (1-34 hours per week), or “reduced hours” (between 25 and 34 hours per week).

We measure three dimensions of time use: paid work, housework, and parenting. “Paid work” measures the number of hours per week spent in paid employment or commuting; “Housework” is a combination of housework, outdoor tasks, and errands; “Parenting” is time spent playing with or caring for own children. We define “Home production” as the sum of housework and parenting.

We are also interested in how time is allocated within the couple. To this end, we define “Female share of home production” as the number of weekly hours a woman spends

⁴The sample was expanded in 2011 with the addition of 2,000 households. We include them in our analysis, but all results are robust to restricting data to the original sample.

in home production divided by the sum of weekly hours she and her male partner spend in home production. “Female share of paid work” is defined analogously. “Female share of total income” is the fraction of total annual household labor income earned by the woman. These “share” variables are only defined for heterosexual couples, and only when we observe the underlying time use and income variable for both partners in the couple. Details on variables definition can be found in Appendix A.

Sample Restrictions We make two sets of sample restrictions: one is individual-based, and one is couple-based.

At the individual level, we focus on people aged 15 to 65. When we study parents (from Section 4 onwards) we only keep parents whose first child was born between 2003 and 2017, in order to observe at least two years before and two years after the first childbirth. For parents, we only keep observations from 4 years before childbirth to 7 years after, in order to have a balanced panel when we restrict to the cohorts close to the Fair Work Act. In the pre-post Fair Work Act comparisons, we exclude the cohort of parents who had their first child exactly in 2009, since this cohort is partially treated.

The couple-based restrictions are germane to our study of intra-household task allocation and apply to Sections 4.2.2 and 5.⁵ We focus on cohabiting heterosexual couples, both married and *de facto* partners. This excludes 9% of first births from single mothers and 1% of first births from lesbian couples. Partnerships and marriages don’t last forever: some women who report having a partner the year of their first childbirth no longer do in subsequent years (6% and 11%, 2 and 5 years after first childbirth respectively), and some change partner (0% and 3%, 2 and 5 years after first childbirth respectively). This is not systematically different between couples who have their first child before and after the Fair Work Act. In our primary sample, for each year we match each woman to the partner she indicates that year, regardless of whether he is the partner in the year of her first childbirth. In our secondary sample, we match each woman to the partner she indicates the year of her first childbirth, regardless of whether they are no longer or not yet a couple. Results are practically identical. We only report results on the primary sample, but results on the secondary sample are available upon request.

Since the exposure design relies on a woman’s job before childbirth, for the analysis of Sections 4.2.2 and 5 we restrict the sample to couples in which we can observe it. Namely, we keep women (and their partners) if they work and report an occupation and

⁵We use these sample restrictions also in Section 4.2.2 for comparability of results across the two sections, but all results replicate without these additional restrictions. All of the results in the preceding sections replicate in this smaller sample. All results using these alternative sample definitions are available upon request.

an industry at least once in the five years prior to the birth of their first child. This is not very restrictive: 89% of mothers in our sample satisfy this criterion.

We keep non-parents and parents with older children as controls, meaning we drop individuals who enter the survey with children aged five or younger.⁶ This choice maximizes power while minimizing the risk of negative weights in estimation (Goodman-Bacon, 2021; Sun and Abraham, 2021), since the “treatment effect” of a child stabilizes relatively quickly after childbirth.⁷ In fact, our main results are virtually identical when dropping all individuals who are already parents when they enter the survey, thus keeping only the non-parents as controls.⁸ Our main results are also virtually unchanged when only keeping individuals who become parents in the sample years, thus using only close not-yet-parents as controls (as several authors in the child penalty literature do, e.g. Kleven et al., 2019b). We present here the results using our preferred sample choice, which maximizes power, but the full sets of results using these alternative control groups are available upon request.

Descriptive Statistics The summary statistics in Table 1 show clear gender differences in work characteristics and time use in the Australian population. 72% of women are in the labor force, as opposed to 84% of men, and, conditional on working, women work 31 hours per week on average, while men 42. Conditional on working, 26% of women are in casual contracts and 63% have a permanent contract, while these figures for men are 19% and 72% respectively. Women do the home shift: on average, they spend 31 hours per week in home production, while among men this figure is 18 hours. These aggregates mask extensive heterogeneity by parental status and cohorts, which will be the focus of the rest of the analysis.

⁶In the sample of non-mothers, we assign random fake childbirth years, and we perform symmetrical matching with partner and cleaning procedures as for the mothers.

⁷The childbirth event studies found in the literature, such as those in Kleven, Landais, Posch, Steinhauer, and Zweimüller (2019a) and in Kleven, Landais, and Søgaaard (2019b) for example, generally display marked dynamics in labor market outcomes of mothers in the first few years after their first childbirth, but relatively constant treatment effects after the first child turns four or five. This point is made even more strongly by Kleven (Forthcoming), who states that “The robustness of the results [to controlling for treatment effect heterogeneity] is arguably unsurprising when considering the shape of the baseline event studies. They have two key features: pre-event trends are parallel *and* post-event effects are almost perfectly persistent following a sharp effect at event time zero. In other words, there is virtually no dynamics in the data, except for the sharp effect precisely at event time $\tau = 0$. [...] the absence of dynamics outside $\tau = 0$ indicates that there is little heterogeneity by age at first birth. [...] In general, concerns about the validity of staggered event study designs are warranted primarily in settings with significant dynamics outside the time of the treatment event”, which child penalties in employment outcomes generally do not display.

⁸Keeping only the never parents as controls, and using the Sun and Abraham (2021) estimator as we do in the main event study specifications, eliminates the potential issue of negative weights coming from “forbidden comparisons” and of cross-lags contamination raised, among others, by Sun and Abraham (2021).

3 The Fair Work Act Increased Schedule Regularity for Mothers

In this section, we document – via both qualitative and quantitative evidence – the increase in schedule regularity the 2009 Fair Work Act brought: the probability of new mothers having a regular schedule increased by 45% after the reform. In the next section, we will analyze the implications of this change on maternal labor supply.

3.1 Qualitative Evidence

The Fair Work Act responded to calls for “permanent part-time”: it empowered mothers to reduce hours below full-time without giving up a permanent contract and the regular schedule that comes with it.

3.1.1 The Pre-2009 Status Quo: Less-Than-Full-Time Jobs Were Mostly Casual

Before the Fair Work Act, it used to be uncommon to work less than full-time in permanent contracts, and individuals who wanted to reduce hours – primarily mothers – often turned to casual contracts, characterized by very irregular schedules, or quit.

There are three types of contracts in the Australian labor market. The most common type is permanent (covering 68% of workers in 2019), which does not have a termination date and includes a prescribed number of weekly hours. Fixed-term contracts, although much less prevalent (11%), are very similar to permanent jobs, the only difference being that they have an end date. Finally, a peculiarity of the Australian setting is the existence and widespread use of casual contracts.

Casual contracts are zero-hours contracts with no commitments on either part, without many entitlements, but which pay higher wages. Casual contracts are relatively common in Australia (more than 20% in 2019). They are characterized by the absence of legal entitlements such as paid leave and severance pay; as compensation, casual employees are usually entitled to a pay loading, which was around 20% before 2010, and was set to progressively increase up to 25% (Lass and Wooden, 2019).⁹ Thus, for the average worker, casual contracts are not necessarily inferior to permanent contracts.

However, casual contracts are substantially more likely than permanent to be associated with an irregular schedule, as detailed in Appendix B.1. Even controlling for occupation and industry, workers under permanent contracts are 17 percentage points more likely to

⁹Note that by making casual contracts more attractive, this provision, if anything, plays against the results we find.

have a regular schedule than their casual counterparts, suggesting that unpredictability is a core feature of the contract rather than a byproduct of the industry or of the occupation.

This creates a trade-off, in the pre-Fair Work Act landscape: mothers seeking reduced hours often had to resort to casual, irregular employment, a dynamic Pocock, Skinner, and Ichii (2009) describe as women “accepting casual employment” as the only available avenue for part-time work.

3.1.2 The Wording of the Law and its Interpretation

The Fair Work Act establishes that “An employee who is a parent [...] may request the employer for a change in working arrangements to assist the employee to care for the child if the child [...] is under school age.” Importantly, “The employer may refuse the request only on reasonable business grounds.” Requests that are protected under this provision “include changes in hours of work, changes in patterns of work and changes in location of work,” and, specifically, also “changing from full-time to part-time”.^{10,11}

While the Fair Work Act allows for a generic “change in working arrangements”, it was understood in the media as well as by legal scholars as introducing a right to request “permanent part-time”, namely the option to remain in a permanent contract – with the regular schedule associated with it – while reducing hours below full-time. Note that the law does not *introduce* part-time: in most occupations, workers could reduce hours also beforehand. However, it changes the modality of part-time: while before the Fair Work Act, switching to part-time usually involved moving to a casual contract and an irregular schedule, after the Fair Work Act, parents can switch to part-time while maintaining the permanent contract and regular schedule associated with it.

3.1.3 Qualitative and Preliminary Evidence of the “Success” of the Law

The Fair Work Act increased access to permanent contracts with reduced hours, and thus to the regular schedule associated with it. This shift, emphasized by various scholars, is evident in the language of industry-specific labor regulations, which explicitly began accommodating permanent part-time arrangements after the reform, and in the share of permanent contracts among part-time working mothers, which greatly increased after the reform.

¹⁰Fair Work Act 2009, Section 65, available at this link: <https://www.legislation.gov.au/C2009A00028/2009-07-01/text>; further explanations available at <https://www.fairwork.gov.au/tools-and-resources/best-practice-guides/flexible-working-arrangements>.

¹¹With the Fair Work Amendment Act of 2013, the right to request a change in working arrangements was extended to parents of school-aged children – the flexible specification we employ in our analysis in Section 3.2 accounts for this.

Appendix B.2 provides the details. First, it presents qualitative evidence from legal scholars (Creighton and Stewart, 2010) highlighting how recent legislative reforms have broken the pre-existing link between part-time and casual work. Second, it compares the wording of the awards (industry-specific labor regulations) before 2009, which usually did not consider permanent part-time employment at all, and after 2009, which regulate it in detail. Finally, Appendix Figure B.2 shows that the large shift from casual to permanent contracts among mothers working less than full-time is evident even in the raw data.

3.1.4 Other Parent-Specific Policies

Australian workers have been entitled to 12 months of job-protected unpaid parental leave since 1979.¹² This was extended with the Fair Work Act, which allowed parents to request up to 12 additional months, subject to the employer's approval. This extension had an extremely low take-up – we discuss this change, together with the other policies that were approved as part of the Fair Work Act, in Appendix E.2.

Paid parental leave was signed into law in 2010 (and applies since January 1, 2011), and it allows one of the parents to take up to 18 weeks off from work, paid at the national minimum wage, after the birth of a child. This policy did not substantially change incentives for new parents: the paid leave could not be combined with the Baby Bonus, an unconditional and tax-exempt cash transfer that was available to all families in which new babies were born, which existed since 2002; and the paid leave did not change job protection upon childbirth, which had been a right for Australian workers for three decades already. Consistent with paid parental leave not changing incentives very much, our results are not impacted by its introduction. We describe the reform and address its impact in detail in Appendix F.

Paternity Leave was introduced in Australia in 2013 in the form of 2-weeks at the national minimum wage (Dad and Partner Pay). Its adoption was far from widespread, with limited take-up, both as documented by policy evaluations conducted by the Australian government (Baxter, Budinski, and Stevens, 2024) and as evident in our data. Consistently, our results are unchanged when we restrict the sample to families unaffected by this policy. The policy and its low take-up are described in Appendix E.3; the robustness of our results to the sample of pre-2012 first births is shown in Appendix I.3.

¹²See <https://www.fairwork.gov.au/about-us/workplace-laws/fair-work-system/australias-industrial-relations-timeline>

3.2 Quantitative Evidence: the Fair Work Act Increased Schedule Regularity

We employ a difference-in-differences methodology to show that the Fair Work Act did have quantitative bite: after the passage of the law, mothers of young kids were 28% more likely to report having a regular schedule, with new mothers being particularly affected (45% more likely). This is a large shift in schedule regularity we will leverage in subsequent sections.

3.2.1 Average Treatment Effect

Our starting point is a standard difference-in-difference (event study) design, in which, following the wording of the law, the treated group is parents of children below school age. Later, in a second step, we will test for heterogeneous treatment effects by age of the first child.

Specification We run the following specification, separately by gender:

$$Y_{it} = \alpha \mathbb{T}_{it} + \sum_{j \neq 2009} \beta_j \mathbb{T}_{ij} + \gamma_i + \delta_t + X_{it} + \epsilon_{it} \quad (1)$$

where Y_{it} is the outcome of interest for individual i at time t and $\mathbb{T}_{ij}$ is an indicator function that takes the value of one if i 's oldest child in year j is aged 5 or younger.¹³ The parameters γ_i and δ_t are individual and time fixed effects respectively, and the controls X_{it} are age-of-the-parent fixed effects and dummies for i 's oldest child being in age ranges 6-8, 9-10, older than 10, interacted with year dummies, included to account for potential persistence in parental behavior. The coefficients of interest are the β_j 's, which measure how the behavior of parents whose first child is aged five or younger in year j differs from that of analogous parents in 2009, relative to non-parents.

Threats to Identification The difference-in-differences strategy just described relies on a parallel trends assumption between mothers and non-mothers; namely, that mothers of young children before and after 2009 would have shown similar work arrangements in the absence of the law, relative to non-mothers. We provide supporting evidence for this

¹³In our main specification, treatment is defined based on the age of the first child: this is based on the intuition that time costs of the first child are an order of magnitude bigger than marginal costs for subsequent children (Jones, Karger, and Michelman, 2025). Defining treatment based on the age of the youngest child yields similar (though noisier) estimates, and, when including both, dummies on the age of the youngest child yield coefficient estimates of zero.

assumption in several ways. In addition to showing the absence of pre-trends in Figure 1 discussed below (and related appendix figures), in Appendix D we show that there was no systematic change in fertility or selection into fertility around 2009, that the financial crisis (which was mild in Australia, not even a formal recession) does not play a role in these results, and argue that there was no employer response on the hiring margin (no backlash through decreased hiring).

Result 1: Schedule Regularity Increases for Mothers, Through the Increase of Reduced-hours Permanent Contracts Our main finding in this section is that new mothers are more likely to have a regular schedule after the Fair Work Act than mothers with similarly aged young children before 2009. This is led by an increase in the probability of having a permanent contract – in particular, a permanent contract with reduced hours.

The fraction of mothers of children below school age with a regular schedule increased on average by 9 percentage points (28%) in the years following the Fair Work Act; with the size of the effect increasing over the first few years. From panel (a) in Figure 1, we learn that 32% of working mothers with a first child aged 0-5 in 2009 had a regular schedule, and this figure is similar for mothers of similarly aged children in prior years. However, after 2009, this number started increasing, with a peak increase of 10 percentage points in 2014, and remained higher thereafter.

Consistent with the qualitative evidence presented in Section 3.1, this increase was paralleled by an increase in the probability of having a permanent contract, and in particular a reduced-hours permanent contract (i.e. 25 to 34 hours per week, approximately 80% of full-time). Indeed, panel (b) of Figure 1 shows that while only 7% of working mothers with a pre-school-aged child had a reduced-hours permanent contract in 2009, this figure more than doubled in the following years, with a similar peak increase of 10 percentage points after 3-4 years, and remained at a permanently higher level thereafter. While these estimates condition on working women, panel (c) shows that this pattern holds identically when using the unconditional variable as outcome (i.e. with 0's instead of missing for non-working women), which clarifies that the extensive margin of labor force participation (and the potential ensuing selection) is not driving the observed pattern.

(Null) Result 2: No Change in Other Work Arrangements for Mothers Other work arrangements, while in principle covered by the Fair Work Act, did not change systematically, highlighting the importance of understanding the public discussion around this law rather than simply its words. Flexibility, measured as agreeing with the statement “My working times can be flexible”, did not change (panel d of Figure 1), suggesting that mothers were

not constrained in this dimension before the Fair Work Act; work from home (panel e) did not change either. Consistent with the contemporaneous narrative described in Section 3.1, this evidence supports the claim that the Fair Work Act did have a strong impact on mother’s schedule regularity and predictability, but none on other work arrangements.

(Null) Result 3: No Change in Any Work Arrangements for Fathers We find no systematic changes for fathers of young children around 2009. Panels (f) and (g) of Figure 1 replicate panels (a) and (b) on the sample of men. Permanent contracts are very common among men (80% of working fathers have this arrangement), who tend to work full-time (only 3% are on reduced-hours permanent contracts), and this does not change with the Fair Work Act. For fathers, the probability of being on a regular schedule was almost double that of mothers before the Fair Work Act (62% probability for fathers versus 32% probability for mothers of 0 to 5-year-olds) and was not affected by the law.

3.2.2 Heterogeneous Treatment Effects

We posit that the treatment effect of the law might be heterogeneous depending on the age of the child when the law is passed: a law changing labor market opportunities for women might have a larger effect if passed when a woman’s first child is just born and she is reoptimizing her labor supply, rather than later when certain choices (of leaving the labor market for example) might be hard to reverse.¹⁴

In Appendix C we explicitly account for this potential treatment effect heterogeneity and, indeed, we find that the entirety of the effect of the Fair Work Act on the probability of having a regular schedule and a permanent contract is concentrated among mothers whose first child was born after the passage of the law. In particular, the probability for a mother of a 0-2 year old having a regular schedule increased by 45% after the Fair Work Act, fully driving the effect on mothers of pre-school children documented above. This suggests that the law did not impact mothers whose child was already three or older at the time of passage of the law (despite the fact that the law in principle applies to them too), and mostly benefited mothers who could take advantage of these new work arrangements within a year or two of their first childbirth.

¹⁴Kuka and Shenhav (2024) document this pattern for the US: they find that single mothers exposed to the expansion of the Earned Income Tax Credit immediately after a first birth have 3 to 4 p.p. higher employment in the five years after a first birth than single mothers exposed 3 to 6 years after a first birth.

3.2.3 Interpretation

We interpret the evidence as supporting a few key facts, which inform the subsequent analysis. First, the Fair Work Act had a statistically significant impact only on the work arrangements of mothers, and none on those of fathers. This is consistent with government reports, which indicate that the greatest majority of requests made under the Fair Work Act comes from women (76% of them, according to survey evidence in O’Neill, 2012). This insight, together with the evidence we will present in Section 4.1 of the child penalty being an issue only for mothers, motivates the primary focus on women. Secondly, we have shown that the entirety of the effect is concentrated among women whose first child was born after the passage of the law, and is persistent. This suggests the following pattern: women rearrange their working life and re-optimize soon after they become mothers, based on the constraints they face at that moment, and these arrangements are sticky; so that mothers of three-year-old children, although technically treated by the law, did not change their arrangements when the Fair Work Act was introduced. This supports our choice in the subsequent analysis of defining treatment and exposure to the law based on the year of birth of the first child.

4 Schedule Regularity Increased Maternal Labor Supply

In this section we document that the increase in schedule regularity brought by the Fair Work Act increased hours worked by mothers by 53%.

4.1 The Status Quo: Child Penalties Before The Fair Work Act

We set the stage for the analysis of the impact of the Fair Work Act on maternal time use by presenting the status quo: namely, the evolution of the household’s time use around parenthood in couples who had their first child before 2009.

4.1.1 Measuring the Child Penalty

We define the “child penalty” as the change in an outcome of interest in the seven years following the birth of an individual’s first child, relative to the four years before the child-birth, and relative to a counterfactual trend based on similarly aged childless individuals and individuals with (much) older children. We present it here as an event study. Note that we define the child penalty as the gap between a woman who becomes a mother and her counterfactual self had she not had children, rather than the gap between mothers and

fathers. Using the male-female gap would not alter the conclusions of our analysis, since we do not find any changes around childbirth in most variables for men, neither before nor after the Fair Work Act.

Our main object of interest is how work arrangements and time use change around the birth of their first child for women and men. Formally, we are interested in the coefficients γ_k from the following event study regression:

$$Y_{it} = \sum_{\substack{k=-4, \\ k \neq -2}}^7 \left\{ \gamma_k \times \mathbb{1}\{t - E_i = k\} \right\} + \alpha_i + \delta_t + \beta_{h(i)} + \epsilon_{it}, \quad (2)$$

where E_i represents the year of birth of i 's first child, and α_i , δ_t , and $\beta_{h(i)}$ are individual, time, and age fixed effects respectively. We run this regression separately for men and women, and for couples as a single entity where appropriate. The coefficients of interest, γ_k 's, represent the change in outcome Y k periods after childbirth, relative to two years before – this is the child penalty in outcome Y , k years from childbirth. Never-parents and individuals who don't have a child aged five or younger when they enter the survey are kept in the sample and act as controls.¹⁵ In order to account for heterogeneous treatment effects across cohorts, we estimate the equation above using the Sun and Abraham (2021) estimator.¹⁶ We cluster standard errors at the individual level.

While we don't explicitly account for heterogeneity by age at first birth in our main specification, this is likely to be of second order importance in this setting, and our results are unchanged when we do. Recent papers by Melentyeva and Riedel (2023) and by Adams, Fjællegaard Jensen, and Petrongolo (2024) emphasize that the impact of a child on his or her mother's labor market outcomes depends on the age of the mother upon childbirth, since this correlates with the potential earning trajectories absent childbirth: failing to account for this heterogeneity might introduce bias in the estimation of the child penalty. While this is of first-order importance when the object of the analysis is the child penalty itself, it can be relatively less important when, such as in this case, the main parameter of interest is the *change* in the child penalty following a change in policy, provided that the policy does not impact age at first birth. As we detail in Appendix D (Appendix Table D.1), the Fair Work Act did not impact age at first birth, so we are not worried by heterogeneity by age at first birth biasing our results. In addition, when we do explicitly account for heterogeneity by age at first birth, our estimates are extremely close to those

¹⁵Only keeping never parents as controls yields identical results, as does only keeping close yet-to-be parents as controls (though results are noisier in this smaller sample); footnotes 7 and 8 in Section 2 discuss this in detail. The full set of results is available upon request.

¹⁶At the time of writing, accounting for treatment effect heterogeneity was a novelty relative to the usual practice in the literature on the child penalty, as emphasized by Melentyeva and Riedel (2023).

in which we do not.^{17, 18, 19}

4.1.2 Evidence on the Child Penalty Before the Fair Work Act

For women who had their first child before the Fair Work Act, parenthood was associated with a stark transition from permanent to casual employment and a substantial reallocation of time from paid work to home production; not for men.

Mothers Transition from Permanent to Casual Contracts Before the Fair Work Act, women moved to casual jobs after becoming parents, while men did not. As explained in Section 3.1.1, before the Fair Work Act it used to be very hard to reduce hours below full-time while keeping a permanent job. Instead, in order to reduce hours, people often had to move to casual contracts, which don't require any commitment and are associated with irregular hours. Unsurprisingly, motherhood, with its large increase in time demands, was associated with a sharp transition from permanent to casual jobs. This is evident in panel (a) of Figure 2, which plots the estimate of the child penalty in the probability of being on a permanent contract conditional on working, separately for women (in pink circles) and men (in blue squares). From the figure, we see that, while the pre-birth prevalence of permanent contracts among working men and women was the same (73%), for women we observe a large drop (almost 20 percentage points) in the probability of being on a permanent contract upon motherhood, drop that widens over time to nearly 30 percentage points in the subsequent seven years (the average over this time horizon is 25 p.p.). Recall that this is only among women who are employed, so the drop is not explained by women dropping out of the labor force. Instead, we see no changes in contract type for men who become fathers.

Partners Specialize and This has Long Term Consequences We document a post-childbirth decrease in leisure and sleep of both partners, which hides very heterogeneous changes in time use. Panels (c) and (d) of Figure 2 show the change in time use around the

¹⁷Kleven (Forthcoming) finds an analogous equivalence result in the U.S., concluding that “These results speak against any non-trivial bias from heterogeneous treatment effects [by age at first birth].”

¹⁸Note also that a big part of Melentyeva and Riedel (2023)’s critique hinges on the undesirable properties of the two-way fixed effect estimator in the presence of heterogeneous dynamic treatment effects (that is, forbidden comparisons and contamination, introduced in the staggered event-study literature by Sun and Abraham, 2021), which do not apply to our setting since we use the Sun and Abraham (2021) estimator, and not the two-way fixed effects, in our event study specifications.

¹⁹This robustness is not shown due to space constraints (since the estimation of the child penalty *per se* is not the main objective of the paper), but it is available in an earlier working paper version (Ciasullo and Uccioli, 2025, Appendix E).

birth of their first child, for women and men respectively. Post-childbirth time use diverges sharply by gender: while men's paid and household work change marginally (decreasing by 2.6 and increasing by 1.4 hours/week, respectively), women reduce paid work and commuting by 28 hours (74%) to accommodate 42 hours of parenting and 11 hours of additional housework. Consequently, women's leisure and sleep decrease twice as much as men's. This indicates that the large increase in maternal labor supply that we document in the next section did not stem from a surplus of leisure time.

Choices of time use made when the first child is born have long-lasting consequences. From panel (c) of Figure 2, we see that, while time spent taking care of the child decreases over time, women do not increase time in paid work symmetrically, suggesting that work choices made at a time of high pressure – the arrival of a baby – might be hard to reverse even after the direct pressure decreases. This translates into a permanently lower income for mothers. If there isn't perfect income pooling, or if bargaining power in a couple depends on income, this might have adverse consequences for women's welfare in the long run.

4.2 The Impact of Schedule Regularity on the Child Penalty

Exploiting the quasi-experiment brought about by the Fair Work Act, we show that having access to a job with a permanent contract and reduced hours – and the ensuing regular and predictable schedule – dramatically lowers the child penalty in labor supply for mothers.

We show this using two complementary empirical strategies. The first exploits variation in the timing of childbirth, while the second adds to it variation in exposure to the law at the occupation-by-industry level, defining individual exposure from the job mothers held before childbirth. We include results from both strategies because we view the first strategy, which compares women who become mothers before and after the Fair Work Act, as a transparent and direct way to show that a large change to the aggregate child penalty occurred exactly the year the Fair Work Act was introduced; while the second strategy, which exploits exposure to the specific provision of the law we are analyzing, as a way to control for potential time-varying unobserved confounders and more credibly claim causality.

We find that women who had their first child after the Fair Work Act are 70% less likely to move out of a permanent contract after childbirth and work 53% more hours per week relative to their earlier counterparts. The entirety of the effect is concentrated among women with pre-birth jobs most exposed to the law.

4.2.1 Strategy 1: Early versus Late Cohorts

Here we briefly present the first empirical strategy and its results; further details are outlined in Appendix E.

Empirical Strategy

We compare the child penalties in contractual form and labor supply for early and late cohorts of mothers, where early cohorts are defined as women who had their first child between 2005 and 2008 (before the 2009 Fair Work Act) and late cohorts are women who became mothers between 2010 and 2013. Here we estimate the child penalty for each year relative to childbirth (Equation 2), separately for early and late cohorts of mothers, keeping the non-mothers and mothers of older children as controls in both. We plot the resulting estimates in the same graph (Figure 3), to gauge the difference visually. In Appendix E, we also estimate the child penalty for the two groups of cohorts jointly, in a more succinct specification, and formally test for their difference.

Results

Lower Probability of Transitioning to Casual Employment Post-Fair Work Act cohorts of mothers experienced a substantially lower probability of transitioning to casual employment, consistent with the interpretation of the law and the evidence presented in Section 3. Panel (a) of Figure 3 shows the change in the probability of being on a permanent contract conditional on working around motherhood. The green circles replicate the pink line in Figure 2(a) and show that for pre-Fair Work Act cohorts of mothers this probability dropped by 20 to 30 percentage points, with the magnitude of the drop increasing over time. The red squares show that instead for post-Fair Work Act cohorts of mothers this probability dropped far less (about 70% less): women who became mothers after the Fair Work Act did use their new right to remain in permanent and regular employment, instead of switching into casual employment, when moving to below full-time after childbirth.

Treated Mothers Work Longer Hours This change in type of contract and schedule regularity was accompanied by a significant increase on the extensive and, in particular, on the intensive margin. Panel (b) of Figure 3 shows that, on average, late cohorts of mothers returned to work at a higher rate relative to the earlier cohorts, by eight percentage points. Even more starkly, panel (c) shows that, among working mothers, late cohorts worked on average four more hours per week, thus experiencing a post-childbirth drop in hours worked of 14 hours instead of 18. Including both intensive and extensive margin,

late cohorts of mothers worked post-childbirth 53% more hours than early cohorts. This is a very large change, especially in light of the extensive literature finding public policies aiming at improving maternal labor supply mostly ineffective (Cortés and Pan, 2023; Olivetti and Petrongolo, 2017).²⁰

Threats to Validity and Robustness Checks

The interpretation of our results of this early versus late cohorts comparison as causal relies on the assumption that, absent the Fair Work Act, the late cohorts of mothers would have behaved, after childbirth, in the same way as the early cohorts. We support this assumption by discussing and addressing three sets of concerns. First, we show that women in early and late cohorts were similar on observables before becoming mothers, ruling out selection into childbearing as a concern. Second, we show that the change in the child penalty happened sharply around the 2009 cohort rather than being a slow-moving time trend. Third, we rule out potential confounders such as Paid Parental Leave (women affected by this other reform are not driving our results), Dad and Partner Pay (excluding couples potentially impacted by this reform does not change our results), and other provisions in the Fair Work Act (which were mostly systematizing existing rights rather than adding new ones). These robustness checks are detailed in Appendix E.

4.2.2 Strategy 2: Exposure by Occupation and Industry

Despite the fact that the early versus late cohorts comparison presented above is robust to a series of checks, our results might, in principle, still be driven by something systematically changing for mothers in 2009 we have not accounted for. Here we address this concern by adding a further level of variation: exposure to the law, determined by the characteristics of the job women held before childbirth. We find that indeed the entirety of the effect on contract type (lower likelihood to transition from permanent to casual contract after

²⁰ Differently from other countries studied in the literature (e.g. Denmark in Kleven et al., 2019b, Austria in Kleven, Landais, Posch, Steinhauer, and Zweimüller, 2024, US in Cortés and Pan, 2023), in Australia there is no child penalty in hourly wages, neither before nor after the Fair Work Act. This is consistent with a labor market characterized by highly regulated wages, with a comparatively high national minimum wage, and awards that cover most employees and include tables with pay rates by detailed type of job (Creighton and Stewart, 2010, p. 352; Lass and Wooden, 2019). This is also not in contradiction with the changing composition of contracts: while permanent contracts could, in principle, be associated with higher “status” and higher wages within a job, casual workers are mandated by law to be paid with a 20-25% premium. This is called casual loading, and is part of the minimum wage legislation, summarized here <https://www.fairwork.gov.au/tools-and-resources/fact-sheets/minimum-workplace-entitlements/minimum-wages>. Thus, there is no reason to expect that the post-motherhood transition to casual contracts we observe at baseline, or the changes in this transition brought about by the Fair Work Act, would induce a mechanical increase or decrease in hourly wage.

childbirth for post-Fair Work Act cohorts of mothers) and on labor supply (lower drop in labor supply) is concentrated among mothers in the most exposed jobs. In addition, we zoom into mechanisms and margins of adjustment, highlighting the lack of adverse career consequences from this reform.

A Measure of Occupational Exposure

Our empirical strategy is based on the idea that different jobs are differentially exposed to the Fair Work Act and that the pre-2009 share of casual contracts in a job (“casual prevalence”) can efficiently summarize such exposure.

Intuition We rely on the fact that employers could refuse requests made under the Fair Work Act on “reasonable business grounds”. This implies that the reform did not affect some jobs, in which irregular and casual employment is likely inherent to the nature of the job (e.g. hospitality). Conversely, other jobs only offered permanent contracts even before the law (e.g. public administration), thus the Fair Work Act likely did not affect them either. This creates three categories of jobs: jobs with a high share of casual contracts were unaffected by the law for technological reasons, jobs with a very low share of casual contracts were unaffected because there was no scope for change in the direction of more permanent contracts (and more regular schedules), while an intermediate share of casual contracts indicates that there was scope for improvement, without technological reasons preventing it. Therefore, we conjecture that women in this intermediate group of jobs would be most exposed to the reform.

The Intuition in Practice We define a “job” as an occupation-by-industry cell, and we measure casual prevalence as the share of casual contracts within a job using pre-2009 observations. Occupations and industries are identified at the 2-digit level: in our view, this strikes a good balance between cells that are sufficiently granular that people within the same cell do sufficiently similar jobs, while retaining a sufficient sample size for accurately estimating the level of casual prevalence.²¹ Our measure of casual prevalence is the fraction of all worker-years in a given occupation-by-industry cell that report having a casual contract, restricting to observations prior to 2009.

Terciles of Casual Prevalence We split the jobs (occupation-by-industry cells) in terciles of casual prevalence; Appendix Figure G.1 depicts where occupations and industries fall in

²¹This is also a practical choice, since 2-digit is the highest level of granularity afforded by the general release of the dataset.

this classification. We see that the occupations with the lowest casual prevalence, meaning the occupations characterized by the highest fraction of permanent jobs, are corporate managers and various types of professionals, and public administration is the industry in which the smallest fraction of employees are on a casual contract. At the other end of the spectrum, security guards, salespersons, and workers at hotels and restaurants are those most likely to be on a casual contract. In the middle tercile of casual prevalence, we find teaching professionals, office clerks, and life science professionals, working in education and health and social work. It is reasonable to imagine that full-time permanent teachers coexist with casual teachers called in if a colleague is sick, and similarly for nurses. These occupations and industries have both a large fraction of permanent workers and a sizeable fraction of casuals, meaning that there is both scope for the law to bring a change in the direction of more permanent contracts and regular schedules, and no technological constraints against it: these are the occupations and industries in which we expect the Fair Work Act to have the most bite.

The bottom and middle terciles of casual prevalence include jobs that are quite similar, while the top tercile is fairly different: this is why we will mostly focus on the bottom-versus-middle comparison. In Appendix G we discuss the similarity and differences between the terciles. The main takeaway is that jobs in the bottom and middle terciles are similar in terms of weekly hours of work, pay, and educational requirements, while for the top tercile these figures are much lower. Women in these jobs tend to make similar choices in terms of fertility, too. This is why, in our empirical analysis, we will emphasize the comparison between the bottom and middle tercile, as the bottom tercile provides a more credible counterfactual for the middle tercile than the top.

Empirical Strategy

We group mothers by exposure to the Fair Work Act depending on the casual prevalence of the job they held before childbirth, and we test whether the child penalty evolved differently after 2009 for more and less exposed groups of women.

We do so via three related exercises. Each woman is assigned the tercile of casual prevalence of the modal job she held in the five years prior to childbirth.²² First, we replicate the early versus late cohort comparison of Section 4.2.1 separately by tercile, to show that the observed changes are concentrated in the middle (most exposed) tercile. Second, we estimate the time series of child penalties separately by tercile, to show parallel pre-trends in the child penalty between more and less exposed terciles. This is carried out

²²We do the same for the controls (non-mothers and mothers who enter the sample with old children) based on a randomly assigned fake childbirth year.

in Appendix I.1. Third, we estimate the relative difference-in-differences for the child penalties (triple-difference), as explained below.

Estimating the difference-in-differences for the child penalties is a little more involved than a standard diff-in-diff because we are not interested in estimating the treatment effect on Y (e.g. labor force participation), but on the pre-post birth *change* in Y (e.g. the child penalty in labor force participation). In practice, we estimate the following specification:

$$\begin{aligned}
Y_{it} = & \gamma_a \times \mathbb{D}_{it} + \\
& + \gamma_{a,2} \times \mathbb{D}_{it} \times \mathbb{1}\{tercile(i) = 2\} + \\
& + \gamma_{a,3} \times \mathbb{D}_{it} \times \mathbb{1}\{tercile(i) = 3\} + \\
& + \gamma_b \times \mathbb{D}_{it} \times Post(i) + \\
& + \gamma_{b,2} \times \mathbb{D}_{it} \times Post(i) \times \mathbb{1}\{tercile(i) = 2\} + \\
& + \gamma_{b,3} \times \mathbb{D}_{it} \times Post(i) \times \mathbb{1}\{tercile(i) = 3\} + \\
& + \alpha_i + \sum_{d \in \{1,2,3\}} \{\delta_{t,d} + \beta_{h(i),d}\} \times \mathbb{1}\{tercile(i) = d\} + \epsilon_{it}
\end{aligned} \tag{3}$$

where $\mathbb{D}_{it} = \mathbb{1}\{(t - E_i) \in [0, 7]\}$ is an indicator function that takes value of one if i 's first child is aged 0 to 7; the dummy $Post(i)$ indicates that i 's first child was born on or after 2010 (and hence i belongs to the late cohorts, using the terminology from previous sections); $tercile(i)$ is the tercile of casual prevalence of i 's pre-birth job; α_i is the individual fixed effect; and the last line specifies that time and age fixed effects are estimated separately by tercile. Here γ_a is the baseline child penalty for the bottom tercile of casual prevalence, meaning that it is the difference in Y in the seven years following the birth of their first child relative to the four years before, for individuals in the bottom tercile of casual prevalence and who gave birth before 2009, which we take as reference group. $\gamma_{a,2}$ is the baseline difference in the child penalty for the treated group (middle tercile) relative to the reference group (bottom tercile), and $\gamma_{a,3}$ is analogous for the placebo group (top tercile); these baselines are estimated off the early cohorts, meaning mothers who had their first child between 2003 and 2008. The coefficient γ_b is the difference in child penalty for the late cohorts relative to the early cohorts for the reference group, and $\gamma_{b,2}$ is the difference-in-differences coefficient of interest, which captures the difference in child penalty for late versus early cohorts of mothers in the treated group, relative to the same difference for the reference group. $\gamma_{b,3}$ is the difference-in-differences coefficient for the placebo group.

Results

We confirm that the aggregate results presented earlier are concentrated among mothers in the jobs most exposed to the Fair Work Act, supporting the hypothesis that the greater likelihood of being on a permanent contract and on a regular schedule brought by the Fair Work Act was responsible for the observed increase in maternal labor supply (decrease in the child penalty). We also paint a fuller picture of margins of adjustment, highlighting that this law does not draw mothers away from full-time employment nor does it have negative implications for mothers' careers.

Figure 4 replicates Figure 3 (early versus late cohort comparison) but split by terciles of exposure. As in Figure 3, the estimates for the early cohorts of mothers (that is, women who became mothers between 2005 and 2008, before the Fair Work Act) are represented by circles connected by dashed lines, while the ones for the late cohorts of mothers (women whose first childbirth was between 2010 and 2013) are represented by squares connected by solid lines. The panels on the left, in blue, are estimated on women with pre-birth jobs in the bottom tercile of casual prevalence: these women were in jobs where it was already the norm for everyone to be on a permanent contract, and thus were likely not affected by the Fair Work Act. The panels in the middle, in orange, are estimated on women with pre-birth jobs in the middle tercile of casual prevalence: in these jobs, not everyone was already on permanent contracts – and thus there was scope for improvement, but also not everyone was on casual contracts, thus suggesting the absence of “technological” constraints against permanent contracts. Finally, the panels on the right, in grey, are estimated on mothers with pre-birth jobs in the top tercile of casual prevalence: here, casual contracts are presumably part of the “technology” of the job, and it's likely that employers can refuse requests made under the Fair Work Act on “reasonable business grounds”: we do not expect this group to be impacted by the Fair Work Act as much as the middle group (if at all).

Probability of Transitioning Out of a Permanent Contract Upon Childbirth: from 37 Percentage Points to Zero for Most Exposed Mothers Women with pre-birth jobs in the middle tercile of “casual prevalence” experienced a stark decrease in the probability of transitioning to a casual contract after childbirth after the Fair Work Act, while women with jobs in the other two terciles did not. Panel (a) of Figure 4 shows the child penalty in the probability of being on a permanent contract, conditional on working. For women with jobs in the bottom tercile of casual prevalence (blue, on the left), relatively few women transitioned from permanent to casual contract after childbirth at baseline, before the Fair Work Act (9 percentage points); this number is virtually zero for post-Fair Work

Act cohorts of mothers, and the difference between cohorts is not statistically significant (11 p.p., with a standard error of 7 p.p.). Mothers in the top tercile (grey, on the right) experienced a similar decrease in the probability of being in a permanent contract both before and after the Fair Work Act. Women with pre-birth jobs in the middle tercile of casual prevalence – the most exposed – (orange, in the middle) are the ones for which we observe the sharpest change: while the early cohorts experienced a massive drop in the probability of remaining in a permanent contract after childbirth (-37 p.p.), the late cohorts in this group experienced no change in the probability of remaining in a permanent contract after childbirth. The difference is large and significant at the 0.1% level (40 p.p., S.E.: 8 p.p.).

Exposure to Fair Work Act Increased Labor Supply These are the same women for whom we observe the large increase in labor supply. By looking at Figure 4, it is evident that the middle tercile (in orange), which experienced the largest relative increase in the probability of being on a permanent contract (panel a), is also the same group that experienced the largest relative increase in labor supply (panel b, +11 hours per week, with a standard error of 2.6 hours/week).

The triple-difference coefficient estimates are consistent with the patterns visualized in the graphs. In Table 2, we highlight in blue the coefficients of interest ($\gamma_{b,2}$) from running Specification (3), namely those on the triple interaction between post-childbirth (“Child Penalty 0-7”), post-Fair Work Act cohort (“Post-2009 First Birth”), and pre-birth job in middle tercile of casual prevalence (Tercile = 2). They convey the same message as Figure 4 (column 4 corresponds to panel a and column 8 corresponds to panel b); numbers can be slightly different because this specification uses all available cohorts (birthing cohorts from 2003 to 2017, excluding 2009) as opposed to focusing only on the cohorts closest to the Fair Work Act.²³

Using this more succinct specification, we show that the conditional result on permanent contracts is not driven by selective exit. If it was driven by a systematic exit of mothers with a greater propensity to move to casual contracts, the *unconditional* probability of holding a permanent contract would not change for treated mothers, given that both exits and casual contracts are coded as zero in the unconditional variable. Instead, the triple-difference coefficient of interest for the unconditional probability of having a permanent contract (column 2 of Table 2) is slightly larger than the corresponding coefficient on the

²³In the event study graphs we restrict to cohorts 2005-2008 and 2010-2013 in order to have a balanced panel for both early and late cohorts of parents.

conditional probability (column 4), thus ruling out selection through exit as the driver of the observed increase in the conditional probability of having a permanent contract.

Importantly, we further show that the increase in the probability of being on a permanent contract and the increase in the probability of working on a regular schedule are basically overlapping for the most exposed mothers. In column (4) of Table 2, the outcome is a dummy variable for having a permanent contract, conditional on working, while in column (6) the outcome is a dummy for working a regular schedule, also conditional on working. The triple-difference coefficients, highlighted in blue, are virtually identical.

Exposure to the Fair Work Act Increased Employment and Hours, But Not Hourly Wages

The increase in labor supply (decrease in the child penalty) for the mothers most exposed to the Fair Work Act came both from the extensive and intensive margin, and was not accompanied by a change in hourly wages, thus translating into a large increase in annual labor earnings. Table 3 reports the coefficient estimates from running the triple-difference specification (3) on labor market outcomes; again, the coefficients of interest ($\gamma_{b,2}$) are highlighted in blue. These estimates show that the large increase in unconditional hours of work for the most exposed mothers in the middle tercile documented above came both from the extensive margin (+11 percentage points, column 2) and from the intensive margin (+3.7 hours per week, column 4). This increase in labor supply was not accompanied by a change in hourly wages - the coefficient of interest in column (6), albeit positive, is very small and not statistically different from zero. As discussed extensively in footnote 20, this is consistent with the high degree of wage regulation in the Australian labor market. Thus, the high and persistent increase in labor supply translates into a large and persistent increase in earnings (+14,890 2021 AUD per year, column 8), corresponding to a 61% increase (41% smaller child penalty) in annual labor earnings for the mothers in pre-birth jobs most exposed to the Fair Work Act.

Exposure to the Fair Work Act Has Noisily Positive, But Not Statistically Significant, Effects on Career Progression

The large increase in labor supply of exposed mothers translates into a suggestive noisy improvement in career progression, though the aggregate numbers are not significantly different from zero. In Appendix H, we study the impact of the Fair Work Act on the probability of being promoted and on the probability of having a supervisory role. For both variables, the average child penalty in the seven years following the birth of their first child is not statistically different for most exposed mothers (Tercile 2) in post-Fair Work Act cohorts relative to earlier cohorts and to less exposed mothers (Appendix Table H.1). On the other hand, Appendix Figure H.1 shows that for the mothers

with pre-birth jobs most exposed to the Fair Work Act there is a significant increase in the probability of being promoted in the year following the first childbirth. In addition, it shows that, for the most exposed group (middle tercile), the progressive shifting away from supervisory roles after childbirth evident in the early (pre-Fair Work Act) cohorts is no longer happening for the post-Fair Work Act cohorts. While the evidence from the event-studies is suggestive, it is extremely noisy and in both cases only significant in a single relative-event year, thus we are hesitant to draw strong conclusions on the positive career impact of the Fair Work Act; but we can rule out significant negative impacts.

The Compliers Would Have Not Worked Full-Time, And The Gap Between Actual and Desired Hours Does Not Close In Appendix H we also address and rule out two concerns: (i) that mothers who would have counterfactually remained full-time are driven by the Fair Work Act into working part-time; and (ii) that the Fair Work Act “simply” allowed mothers to choose a number of work hours closer to their optimum. In principle, a reform that makes less-than-full-time hours less costly (interpreting irregularity as a cost) might also entice women who would have worked full-time to work part-time. In fact, we do not see it happening: there is no decrease in the probability of working full-time; if anything, the point estimates go in the opposite direction (see Appendix Table H.1 and Appendix Figure H.1). Another possible concern is that casual contracts offer suboptimally low hours, and the new reduced-hours permanent contracts simply offer hours that are closer to the optimal number of part-time hours mothers desire. In fact, the gap between actual and desired hours does not close for the exposed group (the point estimates, though insignificant, go in the opposite direction; see Appendix Table H.1). Intuitively, when the only contract available is a casual contract with irregular hours, mothers *want* fewer hours, because each hour is more costly with an irregular schedule. With the regular schedule granted by permanent contracts, mothers want (and get) more hours. So we can rule out the explanation that the permanent contracts increase hours simply by getting mothers closer to their desired level of hours.

Threats to Validity and Robustness Checks

Our job exposure strategy relies on a triple-difference version of the parallel trend assumption: that the *child penalties* of mothers in the treated tercile would have evolved, over time, similarly to the ones of the other two groups. We support this assumption in several ways. First, we show parallel pre-trends in child penalties across terciles. Second, we show no selection into childbirth: early and late cohorts of mothers in the three groups are similar in terms of observables in the pre-birth years, and fertility trends are similar

across terciles. Third, we rule out selection into occupation by showing that our results are robust to assigning women the exposure of the job they had before the Fair Work Act. Fourth, we show that fertility is not affected and that possible differential fertility timing is not driving the results. We discuss these four sets of checks in detail in Appendix I.

5 The Division of Home Production

We do not find any evidence that increased labor supply and earnings of mothers impact how the time burden of children is shared among partners, or that men’s time use is impacted at all.

Here we replicate the analysis of Section 4.2.2 on men’s and couple’s outcomes. In what follows, treatment status for men is defined as their female partner’s treatment status, and treatment status for a couple is defined as the woman’s treatment status. That is, men (and couples) in “bottom tercile” are partners of women with pre-birth jobs in the bottom tercile of casual prevalence, and analogously for the other two groups, regardless of their own job.

Mother’s Increased Labor Supply Decreased Housework, But Not Parenting The large increase in maternal labor supply in the most exposed group corresponds to a decrease in housework, but not in parenting. In Figure 4, we can see that the large increase in labor supply of the most exposed group (panel b, +11 hours per week, S.E.: 2.6 hours/week) corresponds to a smaller decrease in housework (panel d, -5.5 hours per week, S.E.: 2 hours/week). Instead, parenting (panel c) is unchanged (for the most exposed group: +1.5 hours per week, S.E.: 3 hours/week), despite the massive increase in labor supply; showing that mothers exposed to the Fair Work Act did not increase their labor supply at the expense of time spent directly with the child. Consistent with evidence from the US (Bastian and Lochner, 2022), this result suggests that in their optimization, mothers who work more cut first on less valuable forms of home production (housework), thus mitigating concerns that working mothers invest less in their children.

(Lack of) Changes in Men’s Time Use Partners of treated mothers don’t change their labor supply, nor their time spent in home production, despite the large changes in their female partner’s time use. As we can see from panel (a) of Figure 5, men do not change their labor supply around childbirth, and this is true for all groups and cohorts. Panel (c) shows that the same is true for housework, except for a small and not significant increase in housework for the late cohorts of fathers in the middle tercile. In panel (b) we see a small

increase in parenting for partners of most exposed mothers (middle panel, in orange), but the increase is not statistically significant and does not significantly alter how home production is shared between the partners, to which we turn next. Consistently, the triple-difference coefficient estimates reported in blue in Table 4 (columns 2, 4, and 6) show no significant change in time use for the partners of most exposed mothers.

Women Contribute More to Household Income But No Less to Home Production The most exposed mothers, who increase their labor supply (thanks to greater access to regular schedules granted by the Fair Work Act) increase their contribution to total household income, but this does not lead to a change in the relative distribution of home production. Panel (a) of Figure 6 displays the child penalty in “Female Share of Household Income”, computed as the fraction of total household annual labor income earned by the woman. For all groups and for all cohorts we observe a decrease in the female contribution to total household income after childbirth, consistent with the post-childbirth decrease in labor supply for women but not for men documented previously. However, for the middle tercile (most exposed to the Fair Work Act), the late cohorts experience a drop in this measure half the size that of their earlier counterparts (while we don’t observe similarly sharp differences between the two cohorts for women in bottom and top terciles). This is a direct consequence of the fact that most exposed mothers increased their contribution to the total labor supply of the household, as shown in panel (b).²⁴ Despite the substantial increase in the female share of household income and of the female share in paid work for most exposed mothers, we do not observe a significant decrease in the female share of home production (panel c). Since housework represents a small share of home production when there is a young child to take care of, the decrease in women’s time spent in housework is compatible with the lack of change in the relative distribution of total time in home production.

This lack of reallocation of home production among partners, despite a substantial increase in the female share of household income, suggests that women’s labor market opportunities are not the main reason we observe such an unbalanced allocation of the time burden of children. In other words, task allocation within the household is not simply a linear function of relative labor market earnings. The reason for the observed unequal division of home production lies elsewhere, possibly in gender norms.

²⁴Here the outcome variable is the fraction of female time spent in paid work over total household time spent in paid work: $\frac{\text{Woman's weekly hours in paid work}}{\text{Woman's weekly hours in paid work} + \text{Man's weekly hours in paid work}}$.

6 Conclusion

This paper studies how schedule regularity and predictability can be a crucial tool to increase maternal labor supply, suggesting changing the structure of work as a key element for policies aimed at reducing gender inequalities in the labor market.

Leveraging the variation introduced by the Fair Work Act, which entitled parents of young children to request a change in their work arrangements, we established three sets of facts. First, the Fair Work Act was used by new mothers to reduce their weekly working hours without renouncing their permanent contract, hence maintaining a regular schedule. Second, we showed that, when given the option to work the regular schedule afforded by a permanent contract, mothers were able to work longer hours than what they would have done with the irregular schedule associated with a casual contract – an increase by 66% for the most exposed group. Third, we turned to the effect on intra-household repartition of tasks of the above-mentioned improvement: while the changes in mothers’ working hours imply a shift towards equality in the male- and female-shares of household income, we did not observe any changes in the female disproportionate share of home production.

We conclude the paper opening the way for further research in this area. First, our paper shows the existence of a “child penalty” in work arrangements: in the quest for less-than-full-time work, mothers might accept suboptimal work arrangements, such as irregular schedules. When these work arrangements are allowed to change, labor supply changes as well. Further research is warranted to explore further both which work arrangements are predictive of the size of the child penalty, and what we can learn from the way work arrangements change around childbirth.

Second, one would expect that changing labor supply around childbirth would be part of an overall rearrangement of tasks within the household. We have argued that this is not the case, and that increasing women’s share of household income does not map into a more equal distribution of home production. Exploring why the reallocation did not happen and understanding what in certain cases prevents labor market conditions from impacting the intra-household allocation of tasks seems like a fruitful avenue for future research.

Finally, this paper is silent on the impact of this policy on firms. We can speculate it being weakly positive – firms are allowed to refuse requests on reasonable business grounds – but we are unable to quantify it with the data available at the time of writing (the Linked Employer-Employee Dataset only starts in 2011, after the passage of the 2009 Fair Work Act we study). Quantifying the impact on firms of allowing workers more control over their schedules is a promising, policy-relevant direction for future research.

Disclaimers

This paper uses unit record data from Household, Income and Labour Dynamics in Australia Survey [HILDA] conducted by the Australian Government Department of Social Services (DSS). The findings and views reported in this paper, however, are those of the author[s] and should not be attributed to the Australian Government, DSS, or any of DSS' contractors or partners. DOI: 10.26193/3QRFMZ.

The research reported herein was performed pursuant to grant RDR18000003 from the US Social Security Administration (SSA) funded as part of the Retirement and Disability Research Consortium. The opinions and conclusions expressed are solely those of the author(s) and do not represent the opinions or policy of SSA, any agency of the Federal Government, or NBER. Neither the United States Government nor any agency thereof, nor any of their employees, makes any warranty, express or implied, or assumes any legal liability or responsibility for the accuracy, completeness, or usefulness of the contents of this report. Reference herein to any specific commercial product, process or service by trade name, trademark, manufacturer, or otherwise does not necessarily constitute or imply endorsement, recommendation or favoring by the United States Government or any agency thereof.

Ciasullo was not affiliated with Cornerstone Research at the time of writing. The views expressed herein are solely those of the author, who is responsible for the content, and do not necessarily represent the views of Cornerstone Research.

References

- ADAMS, A., M. BALGOVA, M. QIAN, AND T. WATERS (Forthcoming): "Firm Concentration & Job Design: The Case of Schedule Flexible Work Arrangements," *Review of Economics and Statistics*. [On page 2]
- ADAMS, A., M. FJÆLLEGAARD JENSEN, AND B. PETRONGOLO (2024): "Birth Timing and Spacing: Implications for Parental Leave Dynamics and Child Penalties," *CEPR Discussion Paper No. DP19324*. [On pages 14, 71]
- BANG, M. (2021): "Job Flexibility and Household Labor Supply: Understanding Gender Gaps and the Child Wage Penalty," Working Paper. [On page 3]
- BARRETT, G. (2018): "The labor market in Australia, 2000-2016," *IZA World of Labor*. [On page 50]
- BASTIAN, J. AND L. LOCHNER (2022): "The Earned Income Tax Credit and Maternal Time Use: More Time Working and Less Time with Kids?" *Journal of Labor Economics*, 40, 573–611. [On page 26]
- BAXTER, J., M. BUDINSKI, AND E. STEVENS (2024): "Take-up of Parental Leave Pay and Dad and Partner Pay among Australian parents," Tech. rep. <https://aifs.gov.au/research/research-reports/>

take-parental-leave-pay-and-dad-and-partner-pay-among-australian-parents.
[On pages 9, 59]

BERTRAND, M., E. KAMENICA, AND J. PAN (2015): “Gender Identity and Relative Income within Households,” *The Quarterly Journal of Economics*, 130, 571–614. [On page 3]

BORLAND, J. (2011): “The Australian Labour Market in the 2000s: The Quiet Decade,” in *The Australian Economy in the 2000s, Proceedings of a Conference, Reserve Bank of Australia, Sydney*, 165–218. [On page 50]

BÜTIKOFER, A., S. JENSEN, AND K. G. SALVANES (2018): “The role of parenthood on the gender gap among top earners,” *European Economic Review*, 109, 103–123. [On page 3]

CIASULLO, L. AND M. UCCIOLI (2025): “What Works for Working Couples? Work Arrangements, Maternal Labor Supply, and the Division of Home Production,” *CESifo Working Paper No. 12105*. [On pages 15, 46, 49, 70, 71]

CORTÉS, P. AND J. PAN (2023): “Children and the Remaining Gender Gaps in the Labor Market,” *Journal of Economic Literature*, 61, 1359–1409. [On page 18]

CREIGHTON, B. AND A. STEWART (2010): *Labour Law*, The Federation Press. [On pages 9, 18, 46, 55, 57, 58]

DEPARTMENT OF SOCIAL SERVICES AND MELBOURNE INSTITUTE OF APPLIED ECONOMIC AND SOCIAL RESEARCH (2020): “The Household, Income and Labour Dynamics in Australia (HILDA) Survey, GENERAL RELEASE 19 (Waves 1-19),” <https://doi.org/10.26193/3QRFMZ>, ADA Dataverse, V7. [On page 4]

GANONG, P., P. J. NOEL, C. PATTERSON, J. S. VAVRA, AND A. WEINBERG (2025): “Earnings Instability,” *NBER Working Paper No. 34227*. [On pages 1, 2]

GOLDIN, C. (2014): “A Grand Gender Convergence: Its Last Chapter,” *American Economic Review*, 104, 1091–1119. [On pages 3, 64]

GOLDIN, C. AND L. F. KATZ (2016): “A Most Egalitarian Profession: Pharmacy and the Evolution of a Family-Friendly Occupation,” *Journal of Labor Economics*, 34, 705–746. [On page 1]

GOODMAN-BACON, A. (2021): “Difference-in-differences with variation in treatment timing,” *Journal of Econometrics*, 225, 254–277. [On page 6]

HANCOCK, K., J. LAFORTUNE, AND C. LOW (2025): “Winning the Bread and Baking it Too: Gendered Frictions in the Allocation of Home Production,” *NBER Working Paper No. 33393*. [On page 3]

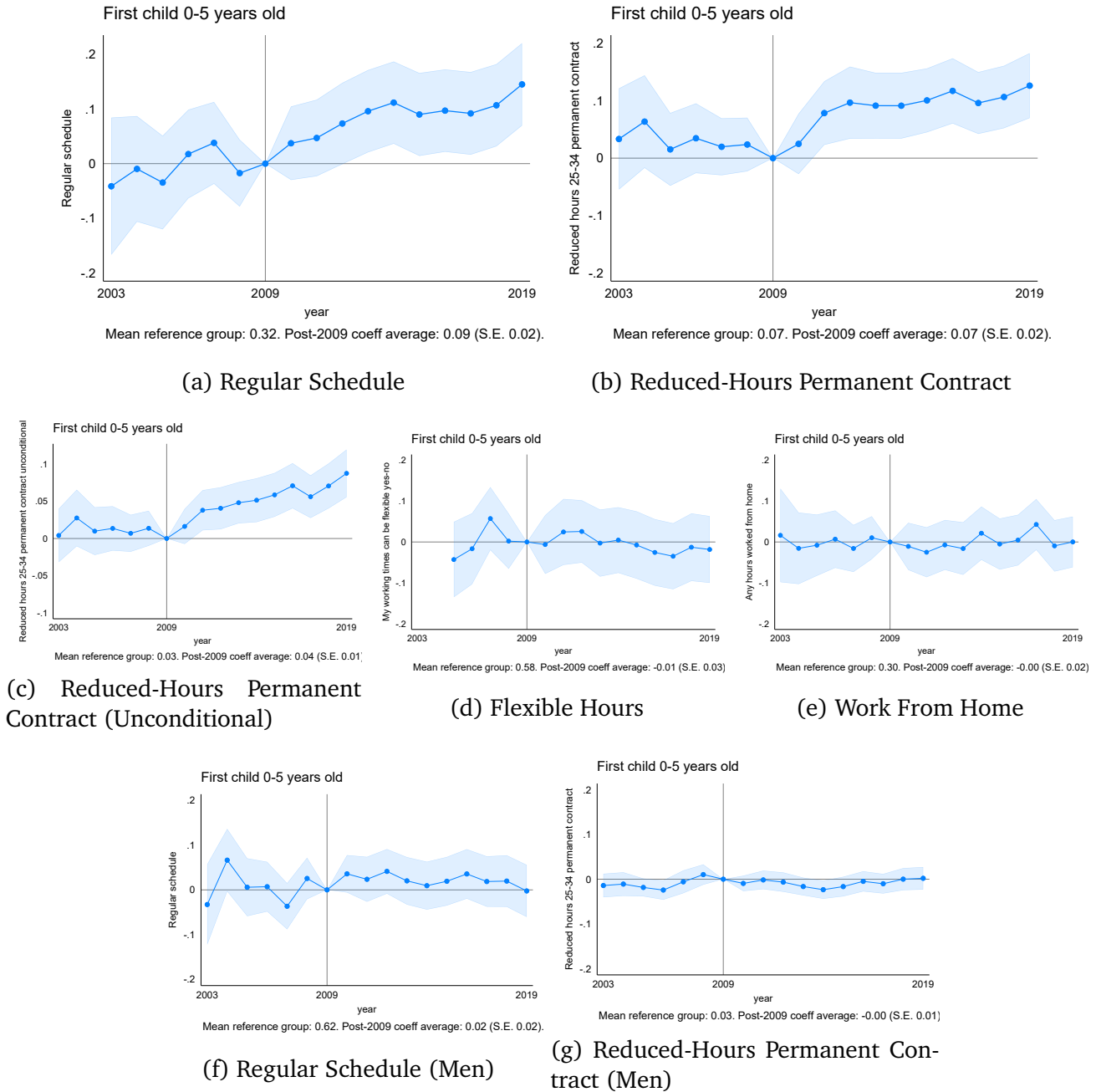
ICHINO, A., M. OLSSON, B. PETRONGOLO, AND P. SKOGMAN THOURSIE (2026): “Taxes, Childcare, and Gender Identity Norms,” *Journal of Labor Economics*, 44, 553–586. [On page 3]

- JESSEN, J., S. SCHWEIGHOFER-KODRITSCH, F. WEINHARDT, AND J. BERKES (2024): “Separate Housework Spheres,” *IZA Discussion Paper No. 17134*. [On page 3]
- JONES, T. R., E. KARGER, AND V. MICHELMAN (2025): “Cheaper by the Dozen: Parental Time Use and Returns to Scale in Raising Children,” . [On page 10]
- KLEVEN, H. (Forthcoming): “The Geography of Child Penalties and Gender Norms: A Pseudo-Event Study Approach,” *American Economic Journal: Applied Economics*. [On pages 6, 15]
- KLEVEN, H., C. LANDAIS, J. POSCH, A. STEINHAUER, AND J. ZWEIMÜLLER (2019a): “Child Penalties across Countries: Evidence and Explanations,” *AEA Papers and Proceedings*, 109, 122–126. [On page 6]
- KLEVEN, H., C. LANDAIS, J. POSCH, A. STEINHAUER, AND J. ZWEIMÜLLER (2024): “Do Family Policies Reduce Gender Inequality? Evidence from 60 Years of Policy Experimentation,” *American Economic Journal: Economic Policy*, 16, 110–149. [On page 18]
- KLEVEN, H., C. LANDAIS, AND J. E. SØGAARD (2019b): “Children and Gender Inequality: Evidence from Denmark,” *American Economic Journal: Applied Economics*, 11, 181–209. [On pages 6, 18]
- KUKA, E. AND N. SHENHAV (2024): “Long-Run Effects of Incentivizing Work after Child-birth,” *American Economic Review*, 114, 1692–1722. [On page 12]
- LASS, I. AND M. WOODEN (2019): “The Structure of the Wage Gap for Temporary Workers: Evidence from Australian Panel Data,” *British Journal of Industrial Relations*, 57, 453–478. [On pages 7, 18]
- MAS, A. AND A. PALLAIS (2017): “Valuing Alternative Work Arrangements,” *American Economic Review*, 107, 3722–59. [On page 3]
- MELENTYEVA, V. AND L. RIEDEL (2023): “Child Penalty Estimation and Mothers’ Age at First Birth,” Working Paper. [On pages 14, 15]
- OLIVETTI, C. AND B. PETRONGOLO (2017): “The Economic Consequences of Family Policies: Lessons from a Century of Legislation in High-Income Countries,” *Journal of Economic Perspectives*, 31, 205–30. [On pages 18, 60]
- O’NEILL, B. (2012): “General Manager’s Report into the operation of the provisions of the National Employment Standards relating to requests for flexible working arrangements and extensions of unpaid parental leave 2009-2012.” Tech. rep., Fair Work Australia. [On pages 13, 56]
- PITTARD, M. (2011): “Reflections on the Commission’s Legacy in Legislated Minimum Standards,” *Journal of Industrial Relations*, 53, 698–717. [On page 55]
- POCOCK, B., N. SKINNER, AND R. ICHII (2009): “Work, Life and Workplace Flexibility: The Australian Work and Life Index 2009,” *Centre for Work+ Life, University of South Australia*, 11. [On page 8]

- RESERVE BANK OF AUSTRALIA (n.d.): “The Global Financial Crisis,” <https://www.rba.gov.au/education/resources/explainers/the-global-financial-crisis.html>, accessed on 03/04/2024. [On page 50]
- SIMINSKI, P. AND R. YETSENGA (2022): “Specialization, Comparative Advantage, and the Sexual Division of Labor,” *Journal of Labor Economics*, 40, 851–887. [On page 3]
- SUN, L. AND S. ABRAHAM (2021): “Estimating dynamic treatment effects in event studies with heterogeneous treatment effects,” *Journal of Econometrics*, 225, 175–199. [On pages 6, 14, 15, 34, 35, 37]
- WATSON, N. AND M. P. WOODEN (2012): “The HILDA Survey: A Case Study in the Design and Development of a Successful Household Panel Study,” *Longitudinal and life course studies*, 3, 369–381. [On page 4]

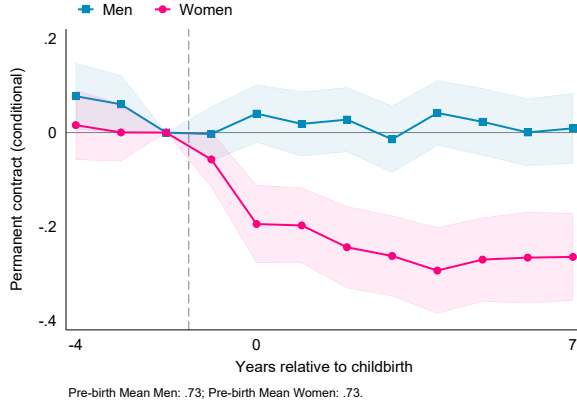
7 Figures

Figure 1: Change in Work Arrangements

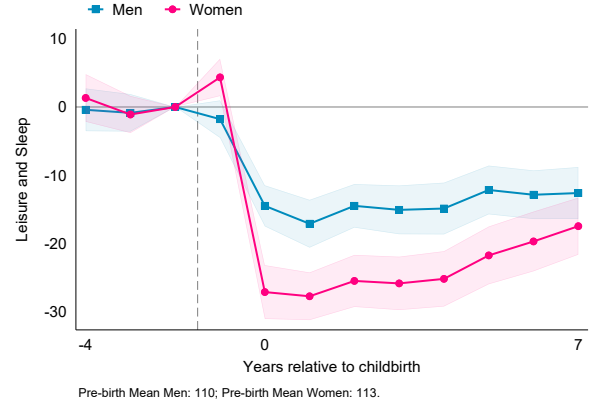


Note: This figure shows the change in work arrangements for mothers relative to non-mothers (panels a through e) and for fathers relative to non-fathers (panels f and g) over time. More formally, we are plotting the estimates of β_j from Equation (1), where the outcome variables are having a regular schedule (panels a and f), having a reduced-hours permanent contract, i.e. a dummy for working between 25 and 34 hours per week under a permanent contract, conditional on working (panels b and g) and unconditionally (panel c), agreeing with the statement “My working times can be flexible” (panel d), working from home at least one hour per week on a regular basis (panel e). See Appendix A for details. Standard errors are clustered at the individual level.

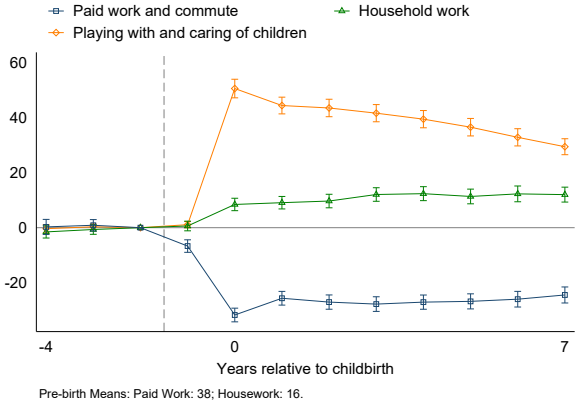
Figure 2: Child Penalties Before the Fair Work Act



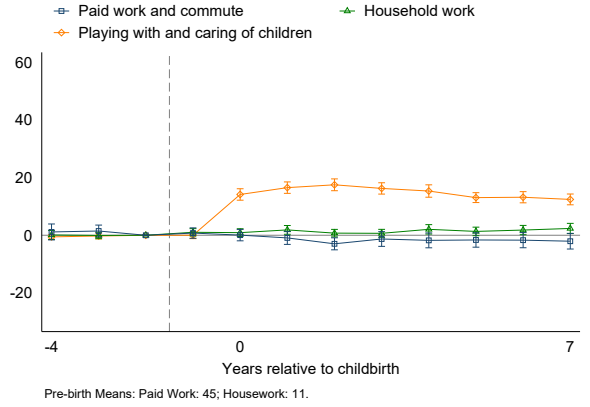
(a) Permanent Contract (Conditional)



(b) Leisure and Sleep [Hours/week]



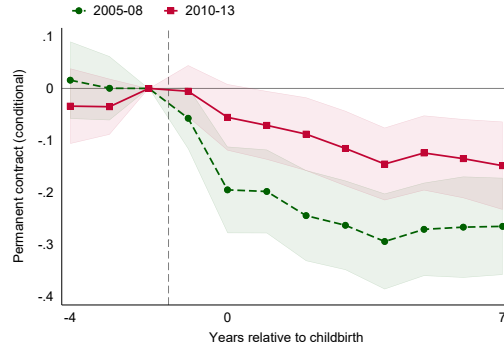
(c) Weekly Time Use - Women



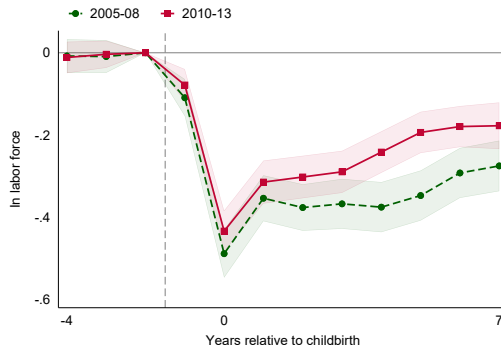
(d) Weekly Time Use - Men

Note: This figure shows the child penalty in the probability of having a permanent contract conditional on working (panel a), in leisure and sleep (panel b), and in different dimensions of time use (panels c and d). Namely, the figure plots the coefficients γ_k 's from estimating Equation (2), which includes individual, time, and age fixed effects. In panels (a) and (b) the blue squares are the estimates for men, the pink circles are the estimates for women, and the shades around them are 95% confidence intervals; the outcomes are, respectively, a dummy for being on a permanent contract conditional on working (panel a) and the number of weekly hours spent in leisure and sleep (panel b). Panels c and d plot the estimates for different dimensions of time use, separately for women (panel c) and men (panel d). The blue squares are estimates for the outcome variable “weekly hours spent in paid work and commuting”, the green triangles for weekly hours spent in housework, and the yellow diamonds for weekly hours spent playing with or caring for the child. We only include parents who had their first child between 2005 and 2008, and we include non-parents and parents of older children as controls - see Section 4.1.1 for details. All regressions are estimated using the Sun and Abraham (2021) estimator, and clustering standard errors at the individual level.

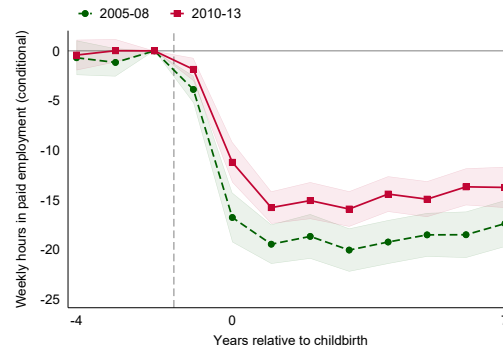
Figure 3: Change in the Child Penalty for Women: Early versus Late Cohorts (Pre- versus Post-Fair Work Act)



(a) Permanent Contract Conditional



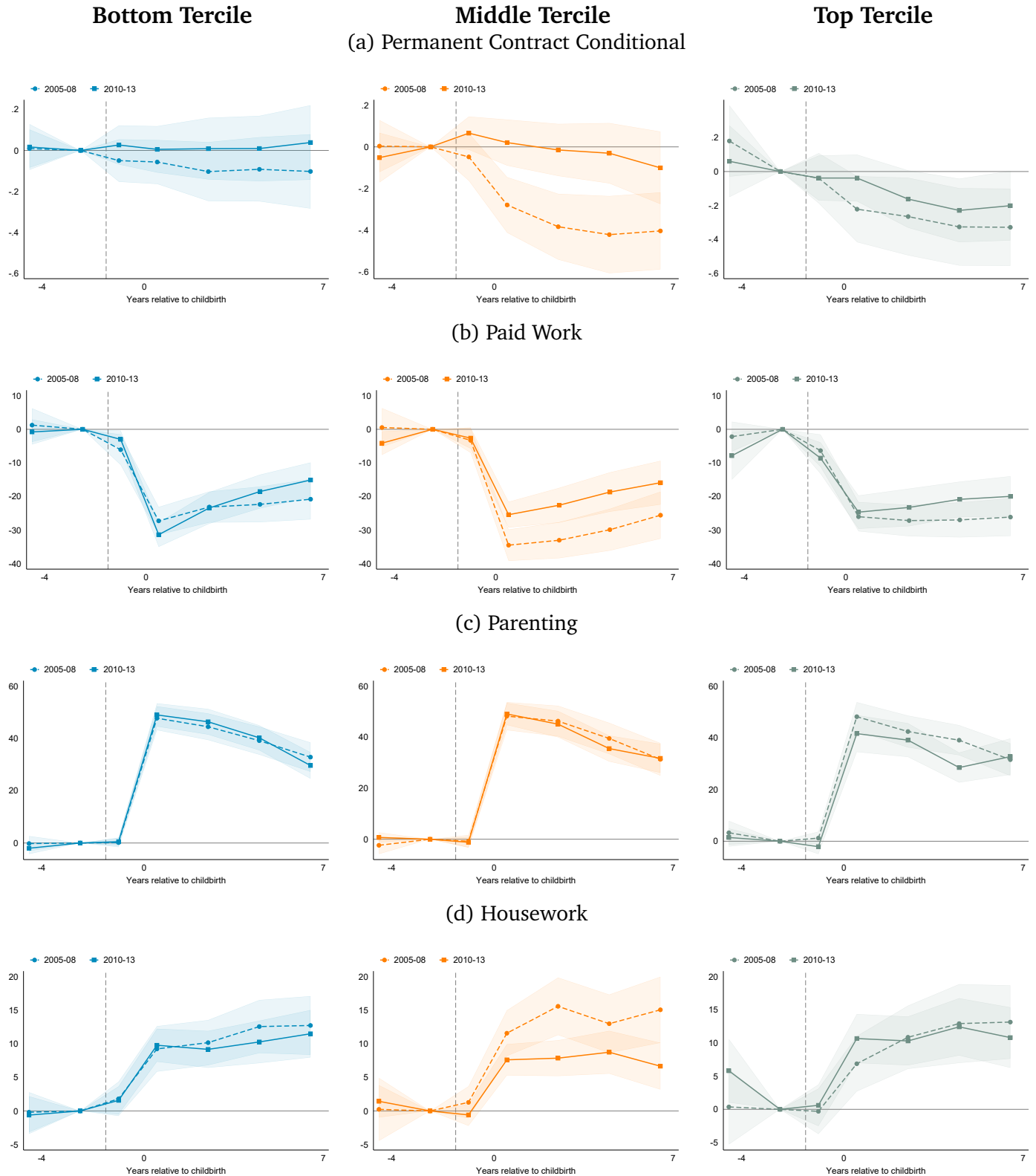
(b) Labor Force Participation



(c) Weekly Working Hours

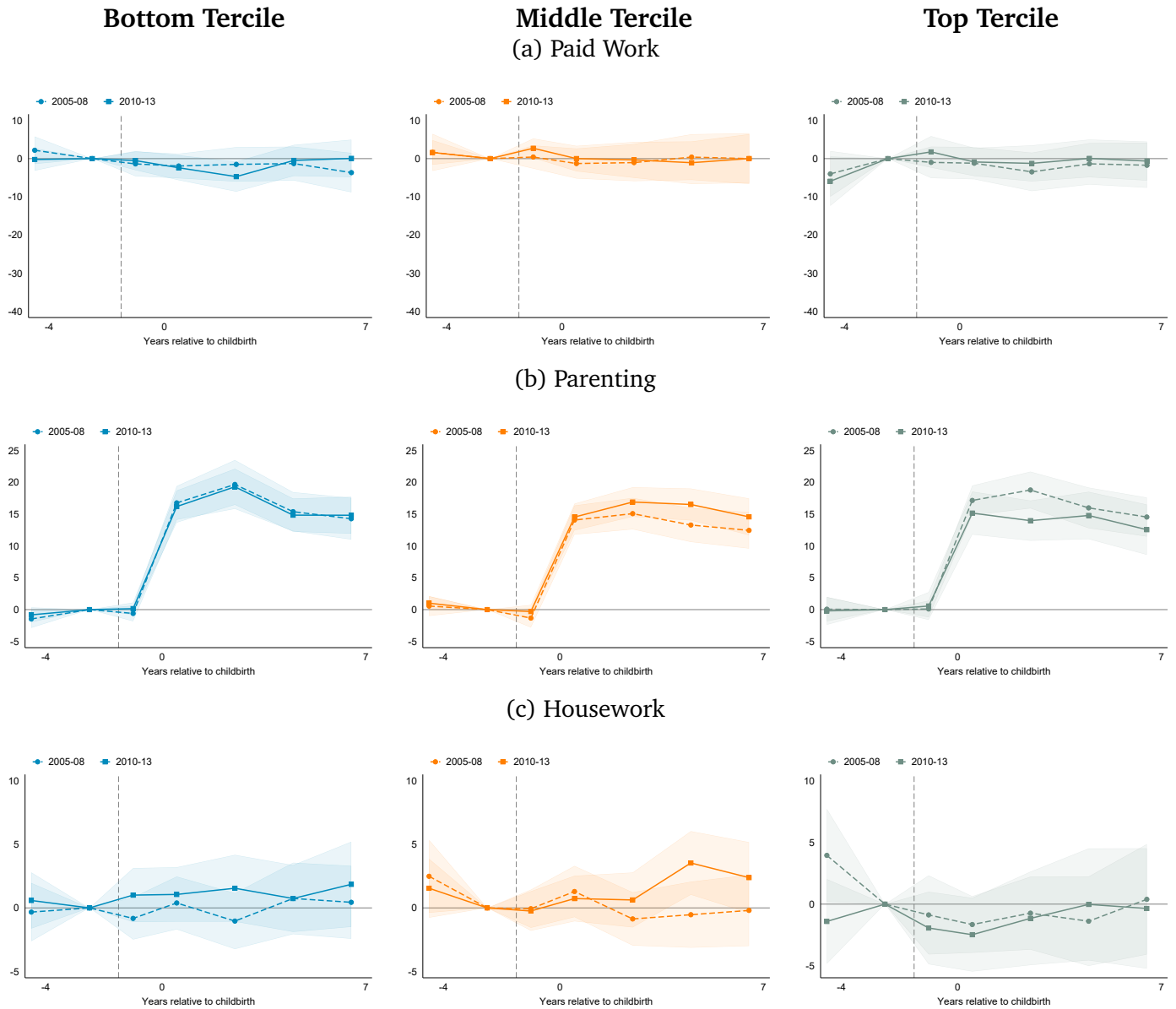
Note: This figure shows how the child penalty in permanent contract and time use for women has changed after the passage of the Fair Work Act. In panel (a) the outcome is a dummy for having a permanent contract conditional on working; in panel (b) the outcome is a dummy for whether the individual is in the labor force; in panel (c) the outcome is weekly hours in paid employment conditional on working. All panels plot the child penalty by year relative to childbirth (the coefficients γ_k 's in Equation (2)) for women, estimated separately for early cohorts (first childbirth between 2005 and 2008, green circles connected by dashed lines) and late cohorts (first childbirth between 2010 and 2013, red squares connected by solid lines). Women without children and with much older children are included in both regressions as controls, and women having children in other sample years are excluded from both regression - see text for details. All regressions are estimated using the Sun and Abraham (2021) estimator, and clustering standard errors at the individual level.

Figure 4: Contract Type and Time Use of Women around Childbirth, Before and After Reform, by Exposure to the Fair Work Act



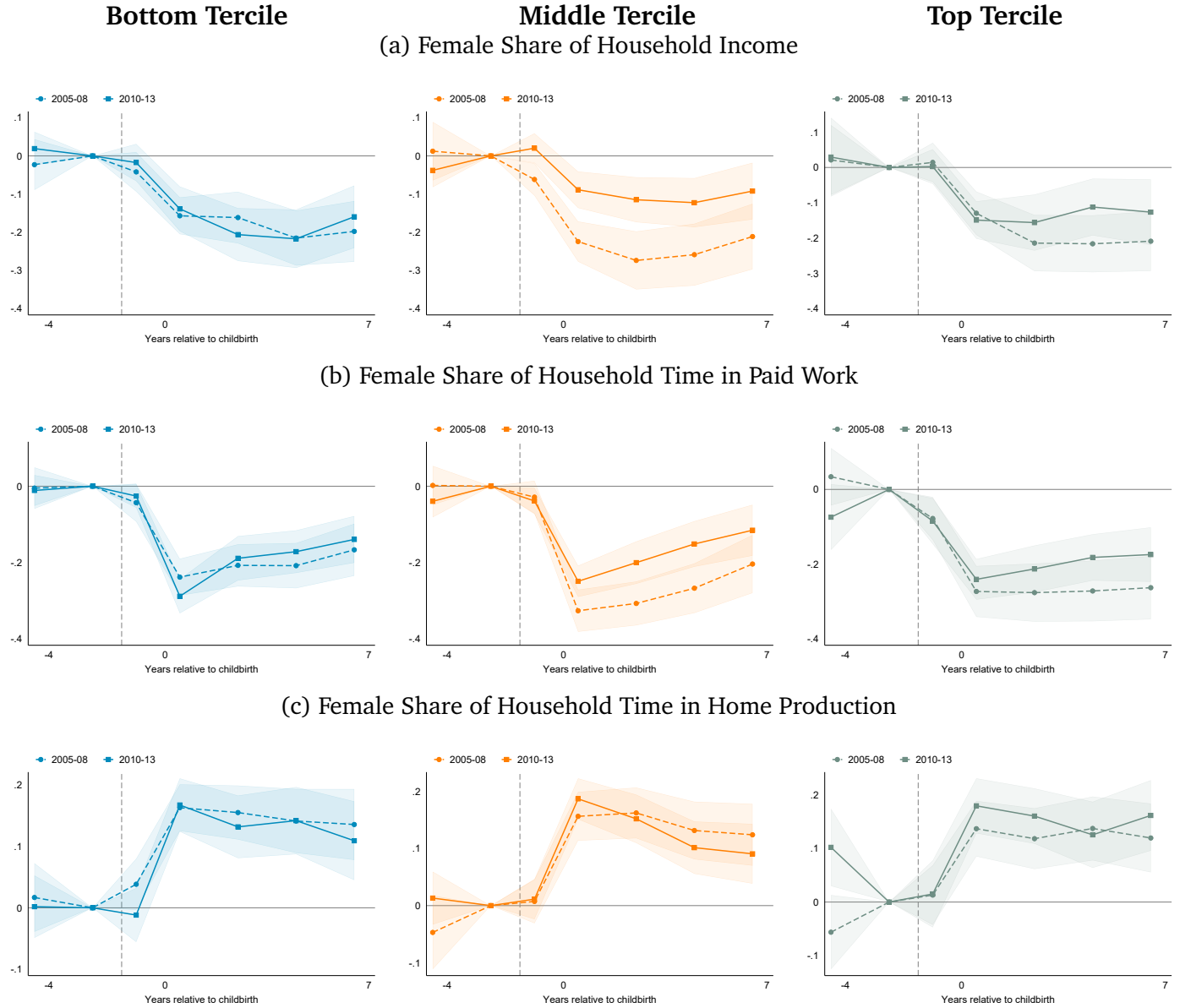
Note: This figure shows the child penalty in permanent contract and time use for women by early and late cohorts and by terciles of casual prevalence. The leftmost panels restrict the sample to women whose modal job in the five years before their first childbirth was in the bottom tercile of casual prevalence (jobs with the highest share of permanent contracts); the middle panels to women whose modal job in the five years before their first childbirth was in the middle tercile of casual prevalence (the most exposed to the Fair Work Act); the rightmost panels to women whose modal job in the five years before their first childbirth was in the top tercile of casual prevalence (jobs with the highest share of casual contracts) - see text for details. Further details on estimation in note to Figure 5.

Figure 5: Time Use of Men around Childbirth, Before and After Reform, by their Partner's Exposure to the Fair Work Act



Note: This figure shows the child penalty in time use for men by early and late cohorts and by their female partner's tercile of casual prevalence; that is, it replicates the time use part of Figure 4 for men, but based on their female partner's treatment status. In each panel, we report estimates of γ_k 's from estimating a version of Equation (2) that pools adjacent years in pairs for power (i.e. the relative-year dummies are bi-annual as opposed to annual). Each set of coefficient estimates comes from separate regressions, all of which include individual, time, and age fixed effects. In the top row, the outcome is weekly hours spent in paid work and commute; in the second row, the outcome is weekly hours spent parenting (playing with and caring for the child); in the bottom row, the outcome is weekly hours spent in household work (housework and errands). Within each panel, the circles connected via dashed lines are estimates restricting the sample to male partners of women who had their first child between 2005 and 2008, while the squares connected via solid lines are coefficient estimates restricting the sample to male partners of women whose first child was born between 2010 and 2013. The shades around the point estimates are 95% confidence intervals. All regressions are estimated using the Sun and Abraham (2021) estimator, and clustering standard errors at the individual level. Further details in note to Figure 4.

Figure 6: Female Share of Household Income, Working Time, and Home Production around Child-birth, Before and After Reform, by Exposure to the Fair Work Act



Note: This figure shows the child penalty in female share of household income, female share in total household time spent in paid work, and female share in total household time spent in home production (sum of housework and parenting) by early and late cohorts and by terciles of casual prevalence. The treatment status of the household is based on the woman's treatment status. See notes to Figures 4 and 5 for details.

8 Tables

Table 1: Descriptive Statistics

	N	Mean	SD	P1	P99
Women					
Age	132421	38.8	14.3	15	65
In labor force	126654	0.72	0.45	0	1
Employed	126654	0.68	0.47	0	1
Weekly hours in paid employment (conditional)	85734	31.4	14.2	3	67
Annual labor income, 2021 AUD (thousands)	132421	35.0	42.1	0	162.9
Hourly wage, 2021 AUD	85734	31.3	32.3	0	99.6
Total time work	106033	23.8	20.6	0	70
Total time housework	106282	21.4	17.3	0	80
Total time parenting	106138	9.61	18.8	0	84
Total time home production	106314	31.0	28.6	0	121
Total time leisure and sleep	104962	109.8	31.2	28	166
Regular schedule	85907	0.50	0.50	0	1
On call and irregular shifts	85933	0.10	0.30	0	1
My working times can be flexible (yes-no)	63065	0.49	0.50	0	1
Any hours worked from home	91716	0.21	0.41	0	1
Permanent contract (conditional)	76380	0.63	0.48	0	1
Casual contract (conditional)	76380	0.26	0.44	0	1
Fixed-term contract (conditional)	76380	0.10	0.30	0	1
Reduced hours (25-34) permanent contract	76380	0.11	0.31	0	1
Has bachelor degree or above	126618	0.27	0.44	0	1
Men					
Age	125761	38.5	14.4	15	65
In labor force	116100	0.84	0.36	0	1
Employed	116100	0.79	0.41	0	1
Weekly hours in paid employment (conditional)	91809	41.7	14.4	5	80
Annual labor income, 2021 AUD (thousands)	125761	60.4	67.4	0	286.2
Hourly wage, 2021 AUD	91809	33.3	28.3	0	116.5
Total time work	93825	37.2	22.7	0	84
Total time housework	94047	13.6	12.0	0	60
Total time parenting	94013	4.60	9.28	0	40
Total time home production	94065	18.2	16.5	0	78
Total time leisure and sleep	93204	110.6	28.5	43	166.5
Regular schedule	91966	0.56	0.50	0	1
On call and irregular shifts	91981	0.10	0.30	0	1
My working times can be flexible (yes-no)	64191	0.52	0.50	0	1
Any hours worked from home	101659	0.22	0.41	0	1
Permanent contract (conditional)	74624	0.72	0.45	0	1
Casual contract (conditional)	74624	0.19	0.39	0	1
Fixed-term contract (conditional)	74624	0.092	0.29	0	1
Reduced hours (25-34) permanent contract	74624	0.021	0.14	0	1
Has bachelor degree or above	116056	0.22	0.41	0	1

Note: The table shows descriptive statistics (number of observations, mean, standard deviation, 1st and 99th percentile) for the variables in the analysis from the full sample, separately by gender. The dataset covers the years 2001-2019. We restrict the sample to people aged 15 to 65. Key variables are defined in Appendix A.

Table 2: Changes in Child Penalty by Exposure to the Fair Work Act - Women (1/2 - Work Arrangements and Time Use)

	(1) Perma- nent contract (uncond)	(2) Perma- nent contract (uncond)	(3) Perma- nent contract (cond)	(4) Perma- nent contract (cond)	(5) Regular schedule	(6) Regular schedule	(7) Total time work	(8) Total time work	(9) Total time home pro- duction	(10) Total time home pro- duction	(11) Total time house- work	(12) Total time house- work
Child Penalty 0-7	-0.369*** (0.0225)	-0.325*** (0.0389)	-0.175*** (0.0293)	-0.0769+ (0.0451)	-0.430*** (0.0285)	-0.380*** (0.0466)	-26.94*** (0.925)	-25.58*** (1.484)	56.92*** (1.325)	55.66*** (1.973)	10.48*** (0.687)	9.080*** (1.093)
* Tercile = 2		-0.122* (0.0528)		-0.174** (0.0645)		-0.135* (0.0658)		-4.927* (2.190)		4.444 (3.068)		2.534 (1.582)
* Tercile = 3		0.0110 (0.0576)		-0.159* (0.0799)		0.00291 (0.0745)		1.513 (2.182)		-0.887 (3.187)		1.425 (1.714)
Child Penalty 0-7 * Post-2009 First Birth	0.152*** (0.0268)	0.0824+ (0.0465)	0.147*** (0.0331)	0.0768 (0.0516)	0.101** (0.0331)	0.0220 (0.0524)	3.742*** (1.057)	-1.032 (1.780)	-1.692 (1.557)	0.479 (2.381)	-2.194** (0.788)	0.276 (1.223)
* Tercile = 2		0.179** (0.0630)		0.152* (0.0739)		0.155* (0.0740)		8.239*** (2.443)		-4.523 (3.517)		-5.113** (1.740)
* Tercile = 3		-0.00110 (0.0687)		0.0387 (0.0885)		0.113 (0.0904)		6.926* (2.702)		-2.678 (4.000)		-1.332 (2.093)
Early (pre-2010) cohorts: Mean Y pre-birth	0.671		0.804		0.724		39.37		16.59		16.01	
- tercile 1		0.768		0.865		0.820		42.58		16.00		15.42
- tercile 2		0.729		0.821		0.713		41.38		15.53		15.07
- tercile 3		0.480		0.687		0.602		33.13		18.54		17.81
Late (post-2010) cohorts: Mean Y pre-birth	0.681		0.771		0.746		40.70		13.53		13.25	
- tercile 1		0.807		0.847		0.860		45.08		13.03		12.86
- tercile 2		0.690		0.756		0.726		41.37		13.37		13.28
- tercile 3		0.484		0.667		0.587		32.78		14.61		13.79
Early cohorts: New parents	415		305		366		396		397		397	
- tercile 1		134		107		128		131		131		131
- tercile 2		138		109		127		132		132		132
- tercile 3		143		89		111		133		134		134
Late cohorts: New parents	734		572		670		695		695		695	
- tercile 1		234		192		223		226		226		226
- tercile 2		296		238		282		283		283		283
- tercile 3		204		142		165		186		186		186
Tot observations	31792	31792	16583	16582	23589	23588	26986	26986	27092	27092	27088	27088

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports the estimates of the child penalty in contract type, work arrangements, and time use for women and its interaction with treatment status (defined by the tercile of casual prevalence of a woman's modal job in the five years before the birth of her first child). More formally, even-numbered columns report coefficients from estimating Specification (3), i.e. γ_a (first row), $\gamma_{a,2}$ (second row), $\gamma_{a,3}$ (third row), γ_b (fourth row), $\gamma_{b,2}$ (fifth row), and $\gamma_{b,3}$ (sixth row). Our main coefficient of interest is $\gamma_{b,2}$ in the fifth row. Odd-numbered columns report estimates from Specification (4), which does not distinguish between terciles of casual prevalence. The outcomes in columns (1) through (6) are dummy variables: in columns (1) and (2), the outcome is an indicator for having a permanent contract (unconditionally), while the same outcome but conditional on employment is in columns (3) and (4). The outcome in columns (5) and (6) is a dummy for having a regular schedule. The outcomes in columns (7) through (12) are measured in hours per week.

Table 3: Changes in Child Penalty by Exposure to the Fair Work Act - Women (2/2 - Labor Market Outcomes)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Employed	Employed	Weekly hours in paid em- ployment (condi- tional)	Weekly hours in paid em- ployment (condi- tional)	Log hourly wage	Log hourly wage	Annual labor income, 2021 AUD (thou- sands)	Annual labor income, 2021 AUD (thou- sands)
Child Penalty 0-7	-0.346*** (0.0191)	-0.291*** (0.0305)	-16.87*** (0.697)	-14.37*** (0.996)	0.0811*** (0.0235)	0.0632+ (0.0362)	-33.13*** (1.588)	-35.26*** (3.289)
* Tercile = 2		-0.0776+ (0.0442)		-5.251*** (1.578)		0.0651 (0.0534)		-3.028 (4.141)
* Tercile = 3		-0.0823+ (0.0475)		-2.312 (1.719)		-0.0409 (0.0597)		10.05* (3.944)
Child Penalty 0-7 * Post-2009 First Birth	0.0749*** (0.0222)	0.00592 (0.0370)	4.294*** (0.797)	2.458* (1.203)	-0.0226 (0.0268)	-0.0445 (0.0441)	7.386*** (1.810)	0.946 (3.732)
* Tercile = 2		0.109* (0.0514)		3.715* (1.796)		0.00800 (0.0627)		14.89** (4.614)
* Tercile = 3		0.0828 (0.0584)		2.671 (2.076)		0.0810 (0.0694)		3.859 (4.583)
Early (pre-2010) cohorts: Mean Y pre-birth	0.923		40.20		2.947		59.73	
- tercile 1		0.951		41.89		2.993		73.85
- tercile 2		0.949		40.87		3.040		64.04
- tercile 3		0.858		36.80		2.743		37.56
Late (post-2010) cohorts: Mean Y pre-birth	0.933		39.20		3.268		62.53	
- tercile 1		0.972		40.81		3.401		78.93
- tercile 2		0.959		39.49		3.267		62.59
- tercile 3		0.828		35.63		3.021		38.54
Early cohorts: New parents	415		365		335		426	
- tercile 1		134		128		121		137
- tercile 2		138		127		117		141
- tercile 3		143		110		97		148
Late cohorts: New parents	734		670		617		782	
- tercile 1		234		222		211		250
- tercile 2		296		283		257		313
- tercile 3		204		165		149		219
Tot observations	31792	31792	23520	23519	19155	19154	35105	35105

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports the estimates of the child penalty in labor market outcomes for women and its interaction with treatment status (defined by the tercile of casual prevalence of a woman's modal job in the five years before the birth of her first child). More details in the note to Table 2. The outcome in columns (1) and (2) is a dummy variable for being in employment; in columns (3) and (4) it is the number of hours of work conditional on being in employment. The outcome in columns (5) and (6) is the logarithm of hourly wages, and in columns (7) and (8) it is annual labor income in thousands of 2021 Australian Dollars.

Table 4: Changes in Child Penalty by Exposure to the Fair Work Act - Men and Couples

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Total time work (men)	Total time work (men)	Total time home pro- duction (men)	Total time home pro- duction (men)	Total time house- work (men)	Total time house- work (men)	Total time home pro- duction (female share)	Total time home pro- duction (female share)	Total time work (female share)	Total time work (female share)	Annual labor income (female share)	Annual labor income (female share)
Child Penalty 0-7	-1.462 ⁺ (0.788)	-1.826 (1.210)	16.38*** (0.770)	18.01*** (1.322)	0.265 (0.441)	1.303 ⁺ (0.778)	0.154*** (0.00890)	0.142*** (0.0144)	-0.259*** (0.0117)	-0.229*** (0.0177)	-0.173*** (0.0122)	-0.139*** (0.0176)
* Tercile = 2		0.218 (1.815)		-3.525* (1.778)		-1.319 (1.020)		0.0303 (0.0210)		-0.0606* (0.0255)		-0.0584* (0.0267)
* Tercile = 3		1.193 (2.000)		-1.394 (2.050)		-1.901 (1.195)		0.00268 (0.0219)		-0.0272 (0.0316)		-0.0410 (0.0317)
Child Penalty 0-7 * Post-2009 First Birth	0.370 (0.878)	0.549 (1.465)	1.017 (0.928)	0.567 (1.604)	0.829 (0.517)	-0.110 (0.938)	-0.000549 (0.0108)	0.0182 (0.0176)	0.0474*** (0.0136)	-0.00814 (0.0215)	0.0588*** (0.0147)	0.0113 (0.0231)
* Tercile = 2		-0.172 (1.981)		1.395 (2.126)		1.066 (1.205)		-0.0313 (0.0249)		0.0888** (0.0295)		0.0925** (0.0324)
* Tercile = 3		-0.572 (2.351)		-0.370 (2.482)		1.963 (1.396)		-0.0298 (0.0276)		0.0878* (0.0372)		0.0463 (0.0395)
Early (pre-2010) cohorts: Mean Y pre-birth	46.82		13.09		12.37		0.551		0.448		0.474	
- tercile 1		49.07		13.22		12.30		0.550		0.454		0.489
- tercile 2		48.51		12.45		12.01		0.553		0.455		0.501
- tercile 3		42.13		13.67		12.86		0.549		0.434		0.423
Late (post-2010) cohorts: Mean Y pre-birth	48.61		12.96		12.30		0.520		0.447		0.453	
- tercile 1		49.27		12.71		12.20		0.518		0.476		0.491
- tercile 2		48.61		13.05		12.44		0.518		0.456		0.456
- tercile 3		47.60		13.16		12.18		0.530		0.383		0.388
Early cohorts: New parents	396		397		397		397		390		391	
- tercile 1		131		131		131		131		130		131
- tercile 2		132		132		132		132		132		130
- tercile 3		133		134		134		134		128		130
Late cohorts: New parents	695		695		695		695		689		688	
- tercile 1		226		226		226		226		225		225
- tercile 2		283		283		283		283		282		280
- tercile 3		186		186		186		186		182		183
Tot observations	26986	26986	27092	27092	27088	27088	27090	27090	25789	25789	25327	25327

Standard errors in parentheses

⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports the estimates of the child penalty in time use for men and in intra-household time allocation and its interaction with treatment status (defined based on the man's female partner). More details in the note to Table 2. The outcomes in column (1) through (6) are measured in weekly hours and are estimated on men only; the outcome in columns (7) and (8) is female share in total household time spent in home production, in columns (9) and (10) it is female share in total household time spent in paid work, in columns (11) and (12) it is female share in annual household labor income.

Online Appendix

List of Appendices

A	Key Variables' Construction	44
B	Characteristics of Casual Contracts and Qualitative Evidence on the Effect of the Fair Work Act	45
C	The Effect of the Fair Work Act by Age of First Child	47
D	Threats to the Identification of the Direct Effect of the Fair Work Act on Work Arrangements	49
E	Strategy 1: Early versus Late Cohorts	52
F	Paid Parental Leave	60
G	Terciles of Casual Prevalence	63
H	Additional Evidence from the Exposure Design	65
I	Threats to Identification in the Exposure Design	69

A Key Variables' Construction

Contract Type and Weekly Hours “Permanent contract (conditional)” is a dummy equal to one if the respondent has a permanent contract, and equal to zero if they have casual or fixed-term contracts (it is missing for individuals not in employment). “Casual contract (conditional)” is similarly defined. “Permanent contract (unconditional)” is defined as its conditional counterpart except that it takes the value of zero (instead of missing) for individuals not in employment. The variable “Reduced hours” is a dummy equal to 1 if the weekly working hours of the respondent are between 25 and 34. “Full-time” refers to working 35 hours per week or more, and “Less-than-full-time” is defined as working between 1 and 34 hours per week.

Regular Schedule and On Call There are mainly two questions that refer to work schedules. The first one, which we label “Regular weekly schedule” asks “On which days of the week do you usually work in your main job?”, and possible answers are “Monday to Friday”, “Nine day fortnight”, that their days vary from week to week, that they vary from month to month, or “other”. If the respondent replies the latter, they are then asked to indicate which days of the week they usually work. The second question is “Regular daily schedule”, and respondents are asked “Which of these best describes your current work schedule in your main job?”. Possible answers are listed below.

1. A regular daytime schedule
2. A regular evening shift
3. A regular night shift
4. A rotating shift (changes from days to evenings to nights)
5. Split shift (two distinct periods each day)
6. On call
7. Irregular schedule
8. Other

Our measure, which is called “Regular schedule” is a dummy variable equal to 1 if the answer to the first question is “Monday to Friday”, or if the respondent indicates four weekdays after answering “other” (e.g. Monday to Thursday), and the answer to the second question is “A regular daytime schedule”, and equal to 0 for any other combination of the two answers. We define someone as being “On Call” if they answer “On Call” or “Irregular Schedule” to the second question.

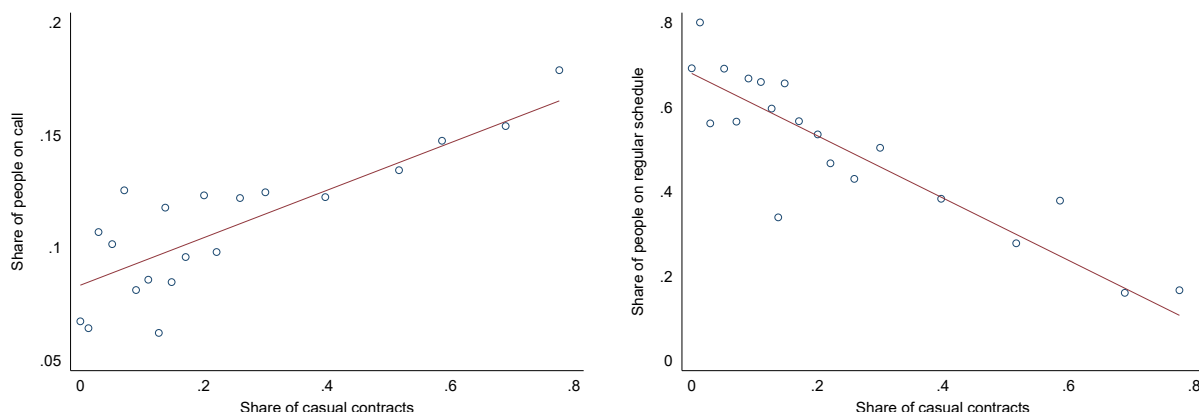
B Characteristics of Casual Contracts and Qualitative Evidence on the Effect of the Fair Work Act

B.1 Casual Contracts Are Associated With Irregular Schedules

Casual contracts are characterized by irregular schedules. Figure B.1 shows the negative correlation between casual contracts and schedule regularity across jobs. In this figure, for each job (2-digit industry-by-occupation cell) we compute the fraction of workers with a casual contract (x-axis, both panels), the share of workers who report being on call (panel a) and the share of workers who report having a regular schedule (panel b), using observation before the Fair Work Act. From these binscatters, the negative correlation between casual contracts and having a regular schedule is very clear across jobs.

Table B.1 shows that this negative correlation between having a casual contract and having a regular schedule (and thus the positive correlation between having a permanent contract and having a regular schedule) also holds within jobs and within people who change contract during their careers: this indicates that this correlation is driven by the nature of the casual/permanent contract itself rather than by other characteristics of jobs it is associated to.

Figure B.1: Correlation between Casual Contract and (Ir)regular Schedule



(a) Share of Casual Contracts and Fraction of Workers on Call (b) Share of Casual Contracts and Fraction of Workers on a Regular Schedule

Note: This figure shows a binscatter between share of casual contracts and two measures of (ir)regular schedule. The unit of observation is a job (occupation-by-industry, 2-digit), weighted by the fraction of workers, all measured before 2009. Variables construction is detailed in Appendix A.

Table B.1: Probability of Having a Regular Schedule by Type of Contract

	(1) Regular schedule	(2) Regular schedule	(3) Regular schedule
Permanent contract (conditional)	0.289*** (0.00263)	0.170*** (0.00256)	0.184*** (0.00278)
Constant	0.362*** (0.00216)	0.440*** (0.00205)	0.436*** (0.00213)
Observations	150955	147482	146227
Included fixed effects	None	Occ-by-ind (2d)	Individual

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table shows the estimated coefficients from a regression of a dummy variable for being on a regular schedule on a dummy variable for having a permanent contract (as opposed to casual or fixed term, conditional on working). Column 1 includes no additional controls, column 2 includes occupation-by-industry (2-digit) fixed effects, and column 3 includes individual fixed effects.

B.2 Qualitative Evidence on the Effect of the Fair Work Act

The advent of “permanent part-time”, meaning the arrangement to work less than full-time but on a permanent contract, is evident both in the wording of the awards and in the raw data, as well as in the discussion of legal scholars.

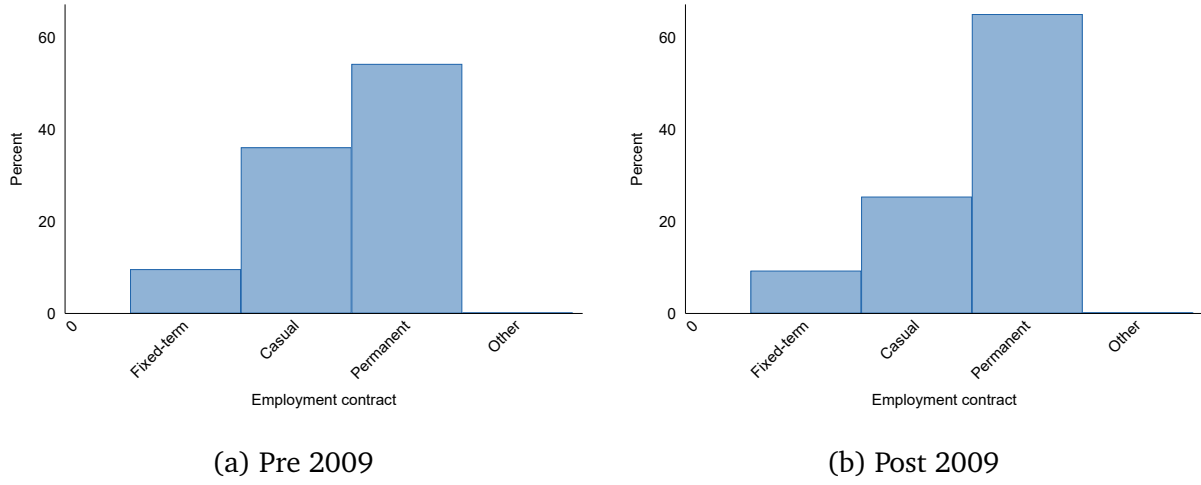
In the edition of their “Labour Law” textbook published right after the passage of the Fair Work Act, Creighton and Stewart (2010) argue that “until recently many awards did not provide for part-time workers to be engaged on anything but a casual basis. But with changing attitudes in the union movement, and legislative reforms, the concept of permanent part-time employment has become well accepted.”

The wording of the awards was changed to introduce and expand on the notion of permanent part-time. For example, the 2010 Modern Award for Private Hospital Employees clearly specifies that “A part-time employee has reasonably predictable hours of work.” and that “Before commencing employment, the employer and employee will agree in writing on a regular pattern of work.” This is in contrast to the corresponding pre-Fair Work Act award, which did not consider regular part-time positions at all.²⁵

Consistent with this narrative around the Fair Work Act, we find suggestive evidence that the Fair Work Act did expand access to permanent contracts with less-than-full-time hours. Figure B.2 shows the type of contract for mothers who work less than 35 hours per week: the large post-2009 increase in the fraction of permanent contracts is evident.

²⁵Further examples are provided in a previous working paper version of this manuscript (Ciasullo and Uccioli, 2025, Appendix Table B.2).

Figure B.2: Type of Contracts among Mothers Working Less Than Full-Time, Before and After the Fair Work Act



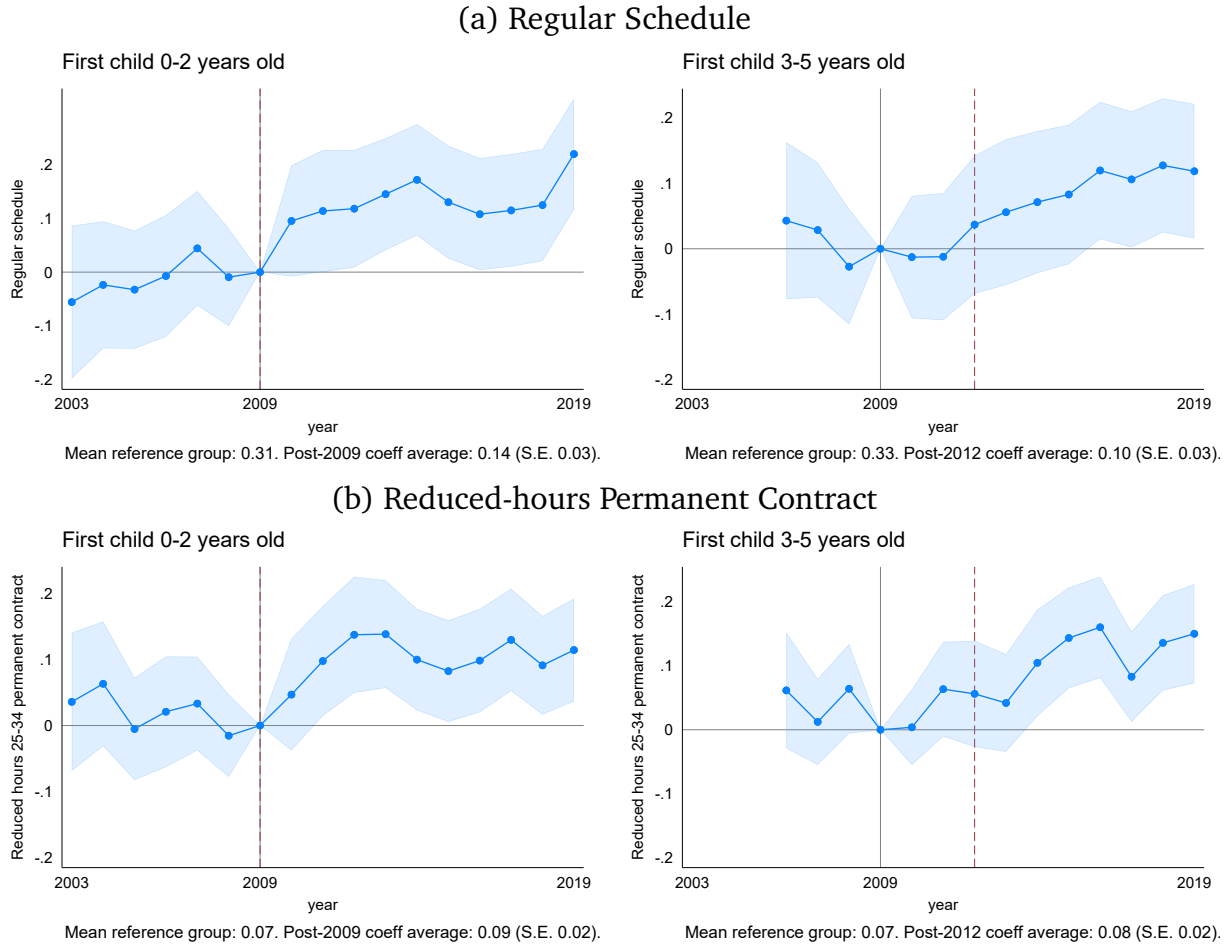
Note: This figure shows the distribution of types of contracts for mothers working less than full-time (1 to 34 hours per week), before and after the 2009 Fair Work Act.

C The Effect of the Fair Work Act by Age of First Child

To support the claim that the change in work arrangements is concentrated among mothers whose first child was born after the Fair Work Act, we show heterogeneity in treatment effects – and in particular, in the timing of treatment effects – by age of the first child.

We test the hypothesis of heterogeneous treatment effects by age of the child by augmenting Equation (1) as follows: we replace $\mathbb{T}_{ij}$, the dummy for i 's first child being aged 0-5, with $\mathbb{T}_{ij}^{(0-2)}$ and $\mathbb{T}_{ij}^{(3-5)}$, dummies for i 's first child being aged 0-2 and 3-5 respectively. Appendix Figure C.1 replicates panels (a) and (b) of Figure 1, on the same outcomes, using this augmented specification. The panels on the left show the coefficient estimates on $\mathbb{T}_{ij}^{(0-2)}$, while the panels on the right show the coefficient estimates on $\mathbb{T}_{ij}^{(3-5)}$. This figure shows that the change in work arrangements happens sharply after 2009 for mothers whose first child is aged 0-2. For mothers whose first child is aged 3-5, the change only happens with a delay, and the uptake of these new work arrangements corresponds to the year in which children born after the Fair Work Act started turning three. This suggests that indeed the law did not impact mothers whose child was already three or older at the time of passage of the law (despite the fact that the law in principle applies to them too), and mostly benefited mothers who could take advantage of these new work arrangements within two years of their first childbirth.

Figure C.1: Change in Work Arrangements for Mothers Relative to Non-Mothers, by Year and Age of Child



Note: This figure shows the change in work arrangements for mothers relative to non-mothers over time. In panel (a) (top row), the outcome is a dummy for having a regular schedule; in panel (b) (bottom row), the outcome variable is the probability of having a reduced-hours permanent contract conditional on working. Relative to the specification in Figure 1, here we replace $\mathbb{T}_{ij}$, the dummy for i 's first child being aged 0-5, with $\mathbb{T}_{ij}^{(0-2)}$ and $\mathbb{T}_{ij}^{(3-5)}$, dummies for i 's first child being aged 0-2 and 3-5 respectively: in this figure we report the relative coefficients $\beta_j^{(0-2)}$ (left panels) and $\beta_j^{(3-5)}$ (right panels). The vertical dashed lines indicate the first cohort that includes a child born after the passage of the law (2009 for 0-2 year old, 2012 for 3-5 year old). Standard errors are clustered at the individual level.

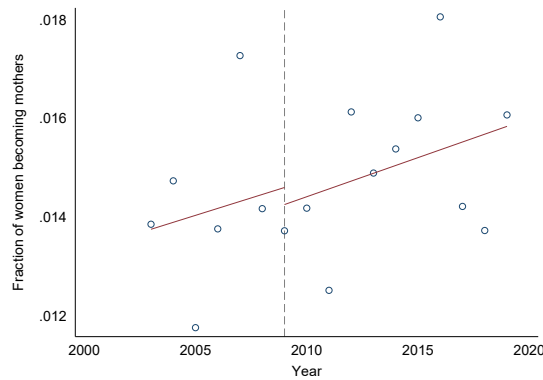
D Threats to the Identification of the Direct Effect of the Fair Work Act on Work Arrangements

Section 3.2 quantifies the effect of the Fair Work Act on work arrangements. These findings rely on the assumption that new mothers of young children before and after 2009 would have shown similar work arrangements in the absence of the law, relative to non-mothers. In this Appendix, we address potential threats to identification. First, we show that fertility does not change systematically around 2009 (Section D.1), and that women who became mothers before and after 2009 are similar on pre-birth observables (Section D.2). Second, we address the concern that a potential “added-worker effect” induced by the global financial crisis could be driving our results, showing that results are robust to controlling for partner’s labor market status (Section D.3). Finally, we document that there was no employer backlash in hiring in response to the reform: after the Fair Work Act, women of childbearing age were not less likely to be employed, did not work for fewer hours, and were not paid less relative to men and older women.²⁶

D.1 No Systematic Changes in Fertility

Australian women did not become mothers at different rates around 2009. Figure D.1 plots the fraction of Australian women in our sample (aged 15-65) who give birth to their first child by year, and superimposes best-fit lines separately before and after 2009. Although there is a slight upward trend in first births, it does not seem to change systematically around 2009, supporting the claim that the law did not directly impact rates of first births.

Figure D.1: Fertility Did Not Change Discontinuously Around 2009



²⁶Not shown here due to space constraints; the full analysis is available upon request, as well as in a previous working paper version of this manuscript (Ciasullo and Uccioli, 2025, Appendix C.4).

D.2 Early and Late Cohorts of Mothers are Similar Pre-Birth

Women who became mothers before and after 2009 were not observationally different on most dimensions when compared before childbirth. This lends support to the hypothesis that they would have behaved similarly also after childbirth, had institutional conditions not changed thanks to the Fair Work Act. Table D.1 presents means of key labor market variables for mothers who had their first child between 2003 and 2008 (early cohorts) and for mothers who had their first child between 2010 and 2017 (late cohorts), measured two years before their first childbirth. The last two columns present the between-groups difference, and standard indications for whether the difference is significantly different from zero. The few significant differences could be by chance (multiple testing) and, if anything, play against the results we find.²⁷

D.3 Accounting for the Financial Crisis

The global financial crisis was particularly mild in Australia, which never went into formal recession (Barrett, 2018; Borland, 2011; Reserve Bank of Australia, n.d.). Thus, it is unlikely to be driving our results: here we show it formally. Table D.2 replicates panels (a) and (b) of Figure 1 in table format, but it also adds controls for whether a woman's male partner is employed and his labor earnings. The coefficient of interest in the first row is practically unchanged from adding these controls. This indicates that the results on the effect of the Fair Work Act on mothers' work arrangements are not driven by women rearranging their labor supply due to their partner's reduced labor earnings from the financial crisis.

²⁷We comment on two variables in particular, for which the difference appears to be significantly different from zero and may raise concerns. First, women in the late cohort have a higher annual labor income on average, compared to women in the early cohort. Because the Australian economy was growing overall, a greater labor income for the late cohort of mothers (measured two years before they give birth) can simply reflect the national trend of increasing wages, rather than selection into fertility. Indeed, the difference in real labor incomes corresponds to an increase by 15.6% between the early and the late cohorts – very close to the 14.2% increase in real annual labor income for all Australian women from the period 2001-2006 to the period 2008-2015. Second, women in the late cohort are less likely to be in a casual contract before childbirth. Even if this was not spurious significance due to multiple hypothesis testing, we argue that this is unlikely to be driving our results. From Table E.1, column 1, we know that for women in the early cohort of mothers the likelihood of leaving their permanent contract is 40 p.p., which, from a baseline level of 59% of women in permanent contract pre-birth (unconditional), implies that 68% of women with permanent contracts in the early cohort leave their permanent contract when they give birth. Given that the fraction of women in permanent contracts pre-birth is slightly larger in the late cohort, the same propensity to leave a permanent contract would translate into a *larger* drop in the probability of holding a permanent contract for the late cohort, when measured in percentage points. If that was the case, we would observe a *negative* coefficient in our specification in Table E.1 (meaning, a larger chance of dropping out from a permanent contract, in percentage points), while we find a positive (and large) one.

Table D.1: Balance Table: Early versus Late Cohorts of Mothers-to-be

	Early cohort	Late cohort	Diff.	SE
Age at first birth	28.72	29.05	0.32	(0.34)
Has bachelor degree or above	0.38	0.39	0.01	(0.03)
In labor force	0.93	0.93	0.00	(0.02)
Weekly hours in paid employment (conditional)	39.18	38.19	-0.99	(0.69)
Annual labor income, 2021 AUD (thousands)	48.64	56.27	7.63**	(2.40)
Hourly wage, 2021 AUD	29.20	30.51	1.31	(0.88)
Total time work	38.18	38.42	0.24	(1.21)
Total time housework	15.73	13.36	-2.37**	(0.86)
Total time parenting	0.50	0.25	-0.25	(0.18)
Regular schedule	0.64	0.69	0.04	(0.03)
On call and irregular shifts	0.06	0.05	-0.01	(0.01)
My working times can be flexible (yes-no)	0.37	0.44	0.07	(0.05)
Permanent contract (conditional)	0.71	0.75	0.04	(0.03)
Casual contract (conditional)	0.18	0.12	-0.06**	(0.02)
Fixed-term contract (conditional)	0.11	0.13	0.03	(0.02)
Reduced hours (25-34) permanent contract	0.06	0.07	0.01	(0.02)

Note: The table shows averages and differences for the sample of will-be mothers. All variables except for age are measured two years before childbirth. Early cohorts include women who had their first childbirth between 2003 and 2008, and late cohorts include women whose first child was born between 2010 and 2017. Stars indicate statistical significance (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Table D.2: Robustness of Effect of Fair Work Act on Work Arrangements to Accounting For The Financial Crisis

	(1) Regular schedule (woman)	(2) Regular schedule (woman)	(3) Reduced hours permanent contract (woman)	(4) Reduced hours permanent contract (woman)
First child 0-5 × Post-2009	0.0709** (0.0252)	0.0715** (0.0252)	0.0763*** (0.0183)	0.0760*** (0.0183)
Employed (man)		0.0290* (0.0134)		-0.0120 (0.0113)
Weekly labor earnings (000) (man)		-0.00864 ⁺ (0.00517)		0.00789* (0.00376)
Pre-period mean	0.328	0.328	0.0885	0.0885
N	31829	31829	27488	27488

Standard errors in parentheses

⁺ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: The table replicates panels (a) and (b) of Figure 1 in table format but includes controls for mothers' male partner's employment status and earnings. Here we restrict the sample to women with male partners.

E Strategy 1: Early versus Late Cohorts

Here we expand on the first empirical strategy we use to study the effect of the Fair Work Act on maternal labor supply. In particular, we estimate the child penalty for the early and late cohorts jointly, in order to formally test for their difference, and we include complementary outcomes in the analysis.

Empirical Strategy We estimate the following model:

$$Y_{it} = \gamma \times \mathbb{D}_{it} + \gamma_{post} \times \mathbb{D}_{it} \times Post(i) + \alpha_i + \delta_t + \beta_{h(i)} + \epsilon_{it} \quad (4)$$

where $\mathbb{D}_{it} \equiv \mathbb{1}\{(t - E_i) \in [0, 7]\}$ is an indicator function for i 's first child being between 0 and 7 years old (which are all the post-childbirth observations we keep for new parents), and $Post(i) \equiv \mathbb{1}\{E_i > 2009\}$ is an indicator function taking value of one if i 's first child was born after 2009 – meaning, an indicator for i being part of the late cohorts. The 2009 cohort is partially treated, and we exclude it from this analysis. Similarly, the year right before childbirth is also partially treated (some women stop working during pregnancy, for example), therefore we also drop it in this analysis. In this specification, the estimate of γ summarizes the child penalty in Y for the early cohorts, while γ_{post} is the difference in child penalty for late cohorts relative to early cohorts, which is the object of interest discussed in this section.

Complementary Results The documented decrease in the child penalty in permanent contracts holds also unconditionally, and it is accompanied by a corresponding decrease in the child penalty in regular schedule. Table E.1 shows that the change in the child penalty in permanent contract is comparable when considering the unconditional probability of having a permanent contract (column 1) as to when using the conditional probability as outcome (column 2) – thus ruling out a systematic exit of mothers who would have moved to a casual contract post-childbirth as the driver of the conditional result. Consistent with the institutional context, the child penalty in the probability of having a regular schedule decreases alongside that of having a permanent contract (column 3), highlighting the strong correlation between contract type and schedule regularity. Note that the increase in the unconditional probability of having a permanent contract is almost twice as large as the increase in employment (column 6), thus showing that the documented intensive-margin changes in work arrangements cannot be fully explained by extensive-margin changes—though greater entry in the labor market because of more suitable work arrangements is part of the story too. Finally, late cohorts of mothers are eight percentage

Table E.1: Child Penalty for Early and Late Cohorts of Mothers

	(1) Perma- nent contract (uncond.)	(2) Perma- nent contract (cond.)	(3) Regular schedule	(4) Weekly hours worked (cond.)	(5) In labor force	(6) Employed	(7) Total time work
Child Penalty 0-7	-0.400*** (0.0245)	-0.252*** (0.0310)	-0.426*** (0.0308)	-18.37*** (0.868)	-0.375*** (0.0223)	-0.362*** (0.0220)	-28.33*** (1.014)
Child Penalty 0-7 * Post-2009 First Birth	0.142*** (0.0335)	0.177*** (0.0394)	0.121** (0.0408)	3.992*** (1.083)	0.0795** (0.0288)	0.0813** (0.0290)	4.610*** (1.292)
Pre-birth mean:							
Early cohorts	0.586	0.727	0.652	38.89	0.920	0.860	37.02
Late cohorts	0.619	0.736	0.677	37.63	0.928	0.873	37.60
Observations:							
N	84594	49993	56229	56116	84594	84594	70618
N Individuals	9625	6858	7369	7368	9625	9625	8638

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports the estimates from running Specification (4): the top row reports the baseline child penalty (γ), while the bottom row reports the change in child penalty (γ_{post}). For consistency with Figure 3, we restrict to four-year cohorts around the Fair Work Act (2005-08 and 2010-13).

points more likely to return to work; the figure is the same both considering labor force participation and employment (columns 5 and 6), indicating that unemployment does not change systematically.

Threats to Identification The interpretation of our results of this early versus late cohorts comparison as causal relies on the assumption that, absent the Fair Work Act, the late cohorts of mothers would have behaved, after childbirth, in the same way as the early cohorts. We support this assumption by discussing and addressing three sets of concerns.

First, we address selection into childbearing. This selection could drive our results if the women who had children after the Fair Work Act are systematically different from their earlier counterparts. As discussed in Appendix D, we find no systematic change in the fertility trend around 2009, nor do we find systematic differences in pre-birth observables between women who became mothers before and after the Fair Work Act.

Second, a simple secular trend in maternal labor supply may explain our results. If the child penalty did not change discontinuously around 2009 but rather slowly evolved, we cannot attribute the observed changes to the Fair Work Act. Section E.1 discusses – and dismisses – this eventuality.

Third, something else may have happened around the same time to explain the discon-

tinuous change in the child penalty we document. First, in Section E.2 we discuss how most of the other provisions of the Fair Work Act were a systematization of pre-existing rights, rather than policies introducing new rights. Second, in Appendix E.3 we show that the two weeks of Paternity Leave introduced four years after the Fair Work Act (in 2013) did not systematically change the amount of leave taken by fathers, and we show that our results are unchanged when we restrict the sample to first births that happened two or more years before this new legislation. Finally, we consider the introduction of federal Paid Parental Leave in Australia in 2010: we show that our results also hold among women not affected (or affected less) by this law, thanks to pre-existing availability of a more generous employer-provided paid leave.²⁸ Appendix F describes this in detail.

E.1 It is Not a Time Trend

So far, we have shown that women who became mothers after the Fair Work Act experienced a systematically smaller child penalty in labor supply and permanent contract relative to their earlier counterparts. Here we confirm that this is driven by a sharp change in 2009 and not by a slow-moving time trend. We do so by estimating the child penalty separately by cohorts, where cohort is now defined by the year of birth of the first child: for all the outcomes of interest, this time series of child penalties is relatively flat before 2009, then shows a discrete jump in 2009, and continues being flat at the new level afterward.

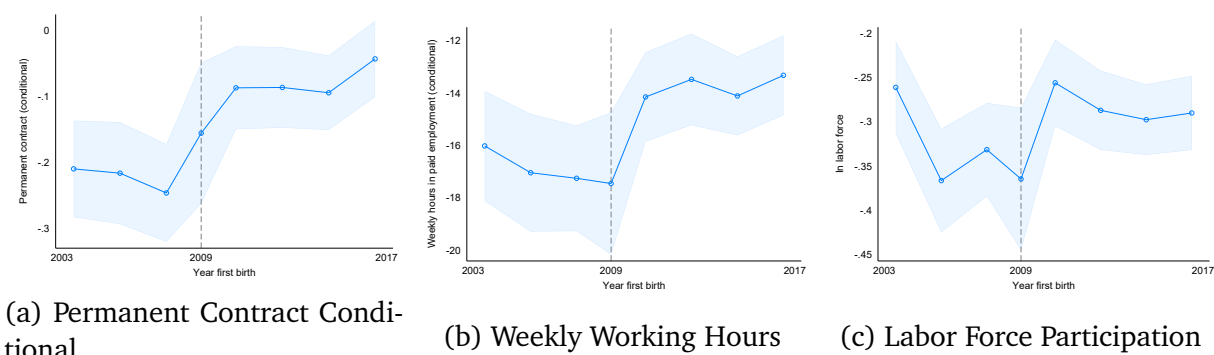
Empirical Strategy We expand on Equation (4) by estimating a separate coefficient by cohort:

$$Y_{it} = \sum_c \left\{ \gamma_c \times \mathbb{D}_{it} \times \mathbb{1}\{c = \text{cohort}(i)\} \right\} + \alpha_i + \delta_t + \beta_{h(i)} + \epsilon_{it} \quad (5)$$

where $\mathbb{D}_{it}$ is a dummy for i 's first child being between 0 and 7 years old (as above) and $\text{cohort}(i)$ is the cohort i belongs to. We define cohorts based on pairs of years of childbirth to preserve statistical power. Here, γ_c is the average child penalty in the first seven years after the birth of the first child, for mothers belonging to cohort c . The sequence of these coefficients shows the time evolution of the child penalty.

²⁸The fact that the formal introduction of Paid Parental Leave is not a confounder in this setting is explained by it not substantially altering pre-existing incentives: as explained in Section 3.1.4 and discussed further in Appendix F, job-protected unpaid parental leave had been available to Australian workers since the 1970s, and new parents had been eligible for a monetary transfer (the Baby Bonus, not combinable with Paid Parental Leave) since 2002.

Figure E.1: Time Evolution of the Child Penalty for Women



Note: This figure shows the child penalty for women (γ_c , on the y-axis) by year of birth of their first child (on the x-axis). The blue circles are the estimates for each biyearly cohort, and the shades around them are 95% confidence intervals. We cluster standard errors at the individual level.

Results Figure E.1 shows that the time series of the child penalties are relatively flat up to 2009, then display a discrete jump around the 2009 cohort, and finally remain flat at a higher level (i.e. smaller child penalties), both for the type of contract and for labor supply, showing that 2009 does indeed represent a discontinuous change in child penalties.

E.2 Other National Employment Standards Provisions

The Fair Work Act established a list of provisions that all awards must include, i.e. minimum employment standards that must be provided to all employees, called National Employment Standards (NES). Many of the provisions in this list were already part of the Australian legislation, and they were included in this list to rationalize or simplify legislation and to have a unique document of minimum standards all employment relations must meet (Creighton and Stewart, 2010, p. 245; Pittard, 2011).

This Appendix lists them and discusses why all of them (except for the provision we study) are either not changing existing rights, or they are quantitatively irrelevant.

The list of National Employment Standards provisions is the following²⁹:

1. maximum weekly hours (38 hours per week; already in place)
2. requests for flexible working arrangements
3. parental leave and related entitlements
4. annual leave (four weeks already in place; extended from four to five weeks in some cases)

²⁹The full text of the Fair Work Act, as approved in 2009, can be found at this link: <https://www.legislation.gov.au/C2009A00028/2009-07-01/text>

5. personal/carer's leave and compassionate leave (10 days per year; already in place)
6. community service leave (introduced with the Fair Work Act)
7. long service leave (already in place)
8. public holidays (already in place)
9. notice of termination and redundancy pay (already in place)
10. Fair Work Information Statement.

The provision we are studying is the second one.

Parental Leave and Related Entitlements (Provision 3) This is the only provision that could, in principle, impact our analysis since it potentially impacts mothers differently than anyone else, but it turns out to be quantitatively irrelevant.

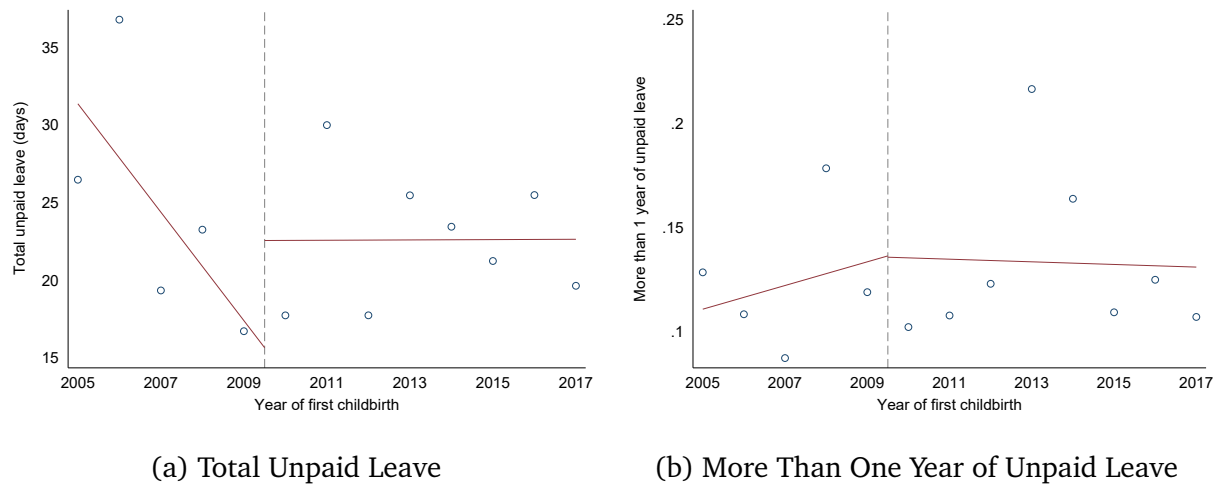
This provision gives workers the right to request up to 12 months of extension to the 12 months of unpaid parental leave they are entitled to, for a total of 24 contiguous months. Note that the entitlement to the first 12 months of job-protected unpaid leave has been available to Australian workers since 1979,³⁰ and was not changed by the Fair Work Act. While the first twelve months are an entitlement, the extension is not: the employee needs to send a request to his or her employer, who has to agree to the extension.

First, we note that the government report that evaluates this aspect of the Fair Work Act (O'Neill, 2012) finds that a very small fraction (less than one-quarter of a percent) of all employees surveyed were considered to have made a request for an extension of unpaid parental leave under the National Employment Standards beyond the 12 months entitlement.

In order to confirm the irrelevance of this legislative change for working mothers in our data, we compute the total number of unpaid leave days taken in the year of first childbirth, and the two subsequent years. Figure E.2 shows the time evolution of the total days of unpaid leave in this three-year window. The left panel shows the total number of days of unpaid leave, and we can see that the average is around 25 days both before and after 2009, far below the maximum leave length that can be taken without the employer's agreement and which was available before the Fair Work Act. The right panel shows the share of women taking more than one year of leave: the numbers are small, and don't change around 2009, further validating the irrelevance, for our results, of the above-described provision.

³⁰The entitlement to the first 12 months of job-protected unpaid leave has been available to Australian workers covered by the award system since 1979, as highlighted in the timeline of Australian industrial relations here: <https://www.fairwork.gov.au/about-us/workplace-laws/fair-work-system/australias-industrial-relations-timeline>

Figure E.2: No Variation in Unpaid Leave Take-Up



Note: This figure plots the average number of days of unpaid leave (panel a) and the probability that leave taken is longer than one year (panel b) in the year of first birth and in the following two years, by year of the birth of the first child.

Maximum Weekly Working Hours and Other Leave Entitlements (Provisions 1,4,5,6,7)

All provisions, except for flexible work arrangements and parental leave (2 and 3), apply to parents and non-parents equally, and therefore should be “controlled for” by our inclusion of non-parents as controls. However, certain provisions might be disproportionately more enticing for working mothers, and may therefore make permanent contracts more attractive for them in particular. In fact, these provisions were largely already in place and were not changed by the Fair Work Act.³¹

³¹As Creighton and Stewart, 2010, (p. 245) write, “Hours of work, parental leave, annual leave and personal, carer’s and compassionate leave were all included in the Standard [section 7 of the Workplace Relations Act 1996].” Specifically:

1. The cap on maximum weekly hours at 38 hours for full-time employees had been in place since 1983 (source: <https://www.fairwork.gov.au/about-us/workplace-laws/fair-work-system/australias-industrial-relations-timeline>), and the Fair Work Act only strengthened the related protections to employees.
4. Annual (vacation) leave was already regulated to be a minimum of four weeks, and the Fair Work Act specifies how it accrues during service.
5. Personal/carer’s leave and compassionate leave is an entitlement of all permanent employees, and it was left unchanged by the Fair Work Act.
6. Community service leave is formally introduced by the Fair Work Act, but was previously regulated in a similar way by industrial instruments or State law, and it covers jury duty or voluntary emergency management activity (i.e. voluntary activity that involves dealing with an emergency or natural disaster).
7. Long service leave is additional time off that accrues to workers if they work for the same employer for a prolonged period of time, and it was left unchanged by the Fair Work Act.

Table E.2: Changes in Other Fair Work Act Provisions for Mothers

	(1) Total days leave	(2) Overtime
Child Penalty 0-7	20.78*** (4.482)	-0.458*** (0.0407)
Child Penalty 0-7 * Post-2009 First Birth	-0.717 (4.881)	0.0668 (0.0502)
Pre-birth mean:		
Early cohorts	21.08	0.656
Late cohorts	25.17	0.585
Observations:		
N	24675	30727
N Individuals	4202	4781

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports the estimates from running Specification (4); see note to Table E.1 for details. The outcomes are: the total number of days of leave in the last 12 months (column 1), and a dummy for working overtime (more than 38 hours per week) conditional on working (column 2). In both cases, the variables are conditional on having a permanent contract. The sample is restricted to women only.

Indeed, we do not observe a significant change in leave-taking and overtime among mothers in permanent contracts after the passage of the Fair Work Act. Specifically, in Table E.2 we show the child penalty in these outcomes among mothers in permanent contracts: late cohorts of mothers do not take more days of leave,³² nor are they systematically less likely to work overtime relative to their pre-Fair Work Act counterparts. This is consistent with the fact that these provisions on the cap of hours and other types of leave were not introducing new rights, and thus did not make permanent contract systematically more attractive to mothers.

Remaining provisions (Provisions 8, 9, 10) Provisions 8, 9, 10 were mostly restatements of existing rights.³³ Provisions 8 and 9 entitled workers to a list of days off corresponding to national holidays and to a written notice in advance of termination, respectively. Provision 10 requires the distribution to new hires of the Fair Work Information Statement, a publication that contains, among others, information about the National Employment Standards, the modern Awards, the role of the Fair Work Act. These provisions

³²Total leave includes: annual (vacation) leave, other forms of paid leave (Maternity; Paternity; Parental; Long-service; Bereavement; Family; Carers; Other paid leave), sick leave, unpaid leave.

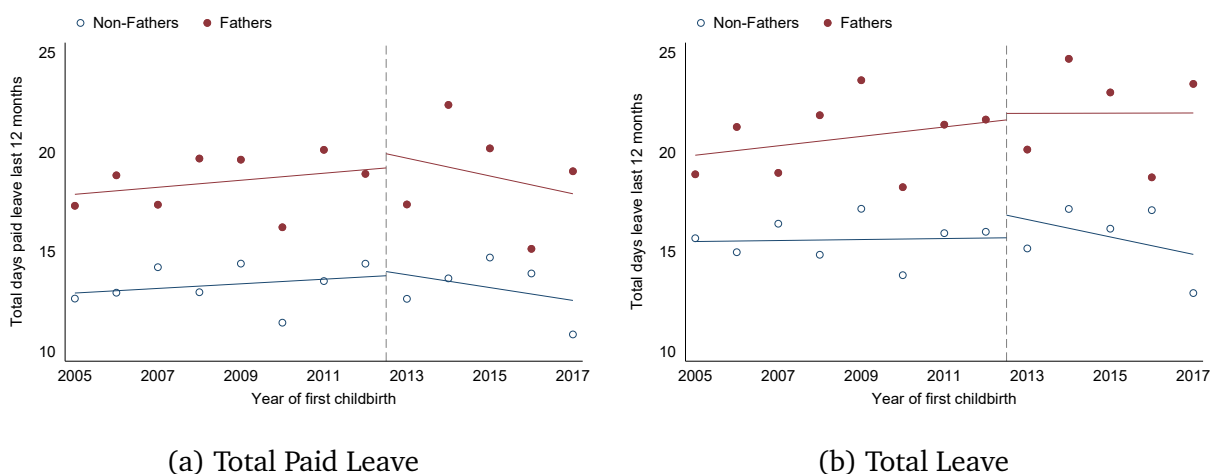
³³Creighton and Stewart (2010, p. 245) explicitly observe that “Public holidays, notice of termination of employment and, after the 2007 amendments, information in the workplace were all dealt with elsewhere in the [Workplace Relations Act 1996].”

cannot be observed directly in our data, but they do not represent changes in mothers' incentives specifically (nor in anyone's incentive, really): therefore, the impact of these provisions, if any, is “controlled for” in all our specifications by the inclusion of non-parents as controls in our analysis.

E.3 Dad and Partner Pay

Australia introduced Paternity Leave (“Dad and Partner Pay”) in 2013, in the form of a 2-week leave paid at the national minimum wage. This policy had limited take-up (40% in 2021, Baxter et al., 2024). In addition, most of it was only replacing other types of leave: Figure E.3 shows how there is no visible change in leave-taking behavior of fathers around the time of the intervention. Indeed, excluding families directly affected by this policy does not change our results: in Appendix I.3, we replicate our main results on the sample of women who became mothers in 2007-08 and 2010-11, well before Dad and Partner Pay. Our main results are consistent, and if anything stronger, on this sample of women whose partner did not have access to paternity leave upon the birth of their first child.

Figure E.3: No Variation in Fathers' Leave-Taking



Note: This figure plots the average number of days of paid leave (panel a) and of total leave (panel b) in the year of first birth, by year of the birth of the first child. For non-fathers, the year of first birth is assigned randomly.

F Paid Parental Leave

In this Appendix, we rule out the possibility that the introduction of a federal Paid Parental Leave in Australia in 2010 may have been the cause of the effects we observe. In order to do this we rely on the fact that approximately half of the Australian workforce already had access to paid parental leave through their employer before the federal reform. Running our analysis on this sample we find unchanged results. The absence of a major effect of Paid Parental Leave on the child penalty in labor supply is consistent with findings in the literature (Olivetti and Petrongolo, 2017), and reasonable in a setting in which Paid Parental Leave replaced a previously existing job-protected unpaid leave and a cash transfer for newborns (thus not dramatically changing incentives for new mothers).

F.1 Institutional Setting

Paid Parental Leave was introduced a year after the Fair Work Act, and could thus act as a confounder. However, the pre-existing presence of unpaid leave with job guarantee for up to one year, and a cash transfer for newborns, implies that this new piece of legislation did not substantially change incentives for new mothers.

Paid parental leave was signed into law in 2010 (and it applies since January 1, 2011), allowing one of the parents to take up to 18 weeks off from work, paid at the national minimum wage, after the birth of a child. The Paid Parental Leave could not be combined with the Baby Bonus, an unconditional and tax-exempt cash transfer that was available to all families who gave birth to or adopted a child. While the Paid Parental Leave is nominally more generous, the interaction with the tax and transfer system meant that for a fraction of families it was less advantageous than the already existing Baby Bonus, and for another set the two were close substitutes. The remaining group, relatively wealthier mothers, were facing a relatively lower replacement rate (since Paid Parental Leave is paid at the minimum wage regardless of pre-birth earnings) and were more likely to be already eligible for employer-provided paid maternity leave, thus it is unclear whether this policy changed their choice set either.

In addition, the reform did not introduce job protection. In fact, up to one year of unpaid job-protected leave upon the birth of a child was already granted to all Australian workers since 1979, so the 2010 reform did not change this aspect. This is crucial because job protection is the aspect of paid leave that really matters for labor force attachment of new mothers, according to the review article by Olivetti and Petrongolo (2017). In conclusion, it seems that this provision did not significantly change the incentives faced by

Table F.1: Child Penalty for Early and Late Cohorts of Mothers

	(1)	(2)	(3)	(4)	(5)	(6)
	Permanent contract (uncond.)	Permanent contract (cond.)	Regular schedule	Weekly hours worked (cond.)	In labor force	Employed
Child Penalty 0-7	-0.417*** (0.0422)	-0.186*** (0.0428)	-0.485*** (0.0502)	-18.69*** (1.277)	-0.313*** (0.0337)	-0.341*** (0.0341)
Child Penalty 0-7 * Post-2009 First Birth	0.171** (0.0550)	0.139** (0.0530)	0.157** (0.0593)	3.786* (1.499)	0.0583 (0.0406)	0.0850* (0.0412)
Pre-birth mean:						
Early cohorts	0.766	0.789	0.744	41.32	0.981	0.977
Late cohorts	0.749	0.796	0.722	39.08	0.980	0.961
Observations:						
N	14677	12376	12831	12814	14677	14677
N Individuals	1069	1057	1066	1066	1069	1069

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table replicates Table E.1 on the subsample of women who, two years before childbirth, were entitled to employer-provided paid parental leave.

new parents, and therefore we do not expect it to lead to significant changes in their labor supply choices.

F.2 Empirical Strategy and Results

In order to study whether Paid Parental Leave affected the change in child penalty we document, we leverage the fact that, before its introduction, 55% of working women were already eligible for employer-provided paid maternity leave. Under the assumption that the introduction of national, state-funded, Paid Parental Leave does not affect women who already can access this benefit through their employer, any change we observe for this group of women can be attributed to the Fair Work Act (and not Paid Parental Leave).

We repeat the analyses of Tables E.1 and 2 on the subsample of women who, two years before childbirth, were entitled to employer-provided paid parental leave.³⁴ Results are presented in Tables F.1 and F.2 respectively: while we lose some statistical power since we are working on a smaller sample (especially for the second table, where we are looking at heterogeneities by exposure to the law), all coefficients have the same sign and similar magnitudes.

³⁴The full set of results estimated on this sample is available upon request.

Table F.2: Changes in Child Penalty by Exposure to the Fair Work Act - Paid Parental Leave Subsample

	(1) Perma- nent contract (uncond)	(2) Perma- nent contract (uncond)	(3) Perma- nent contract (cond)	(4) Perma- nent contract (cond)	(5) Regular schedule	(6) Regular schedule	(7) Total time work	(8) Total time work	(9) Total time home pro- duction	(10) Total time home pro- duction	(11) Total time house- work	(12) Total time house- work
Child Penalty 0-7	-0.355*** (0.0397)	-0.292*** (0.0628)	-0.101* (0.0437)	-0.0156 (0.0595)	-0.472*** (0.0500)	-0.393*** (0.0836)	-31.49*** (1.470)	-28.28*** (2.385)	60.54*** (2.120)	56.94*** (2.903)	11.56*** (1.238)	10.81*** (1.778)
* Tercile = 2		-0.122 (0.0832)		-0.140 (0.0855)		-0.0824 (0.103)		-6.825* (3.130)		6.662 (4.399)		0.632 (2.535)
* Tercile = 3		0.00910 (0.141)		-0.248 (0.191)		-0.343* (0.173)		0.354 (4.306)		4.465 (6.381)		2.131 (4.354)
Child Penalty 0-7 * Post-2009 First Birth	0.149*** (0.0440)	0.0888 (0.0703)	0.102* (0.0460)	0.0479 (0.0653)	0.103* (0.0517)	0.00432 (0.0826)	5.539*** (1.598)	1.681 (2.592)	-3.553 (2.364)	-0.297 (3.270)	-3.195* (1.289)	-0.916 (1.906)
* Tercile = 2		0.157+ (0.0944)		0.0988 (0.0949)		0.120 (0.107)		7.568* (3.429)		-4.893 (4.871)		-4.269 (2.658)
* Tercile = 3		-0.152 (0.159)		0.108 (0.197)		0.331+ (0.179)		0.920 (4.916)		-8.140 (7.356)		-1.503 (4.521)
Early (pre-2010) cohorts: Mean Y pre-birth	0.792		0.832		0.799		45.06		14.86		14.42	
- tercile 1		0.860		0.903		0.900		46.21		16.13		15.08
- tercile 2		0.756		0.783		0.704		44.65		14.77		14.77
- tercile 3		0.700		0.758		0.829		42.40		10.37		10.37
Late (post-2010) cohorts: Mean Y pre-birth	0.765		0.803		0.793		43.53		13.08		12.91	
- tercile 1		0.849		0.861		0.868		45.80		12.90		12.82
- tercile 2		0.737		0.770		0.778		42.98		13.30		13.27
- tercile 3		0.647		0.765		0.642		39.55		12.82		11.90
Early cohorts: New parents	112		100		108		110		110		110	
- tercile 1		44		41		44		44		44		44
- tercile 2		54		48		52		53		53		53
- tercile 3		14		11		12		13		13		13
Late cohorts: New parents	307		263		298		300		300		300	
- tercile 1		112		97		108		110		110		110
- tercile 2		146		127		145		143		143		143
- tercile 3		49		39		45		47		47		47
Tot observations	10141	10138	6553	6551	8199	8196	9101	9098	9127	9124	9126	9123

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

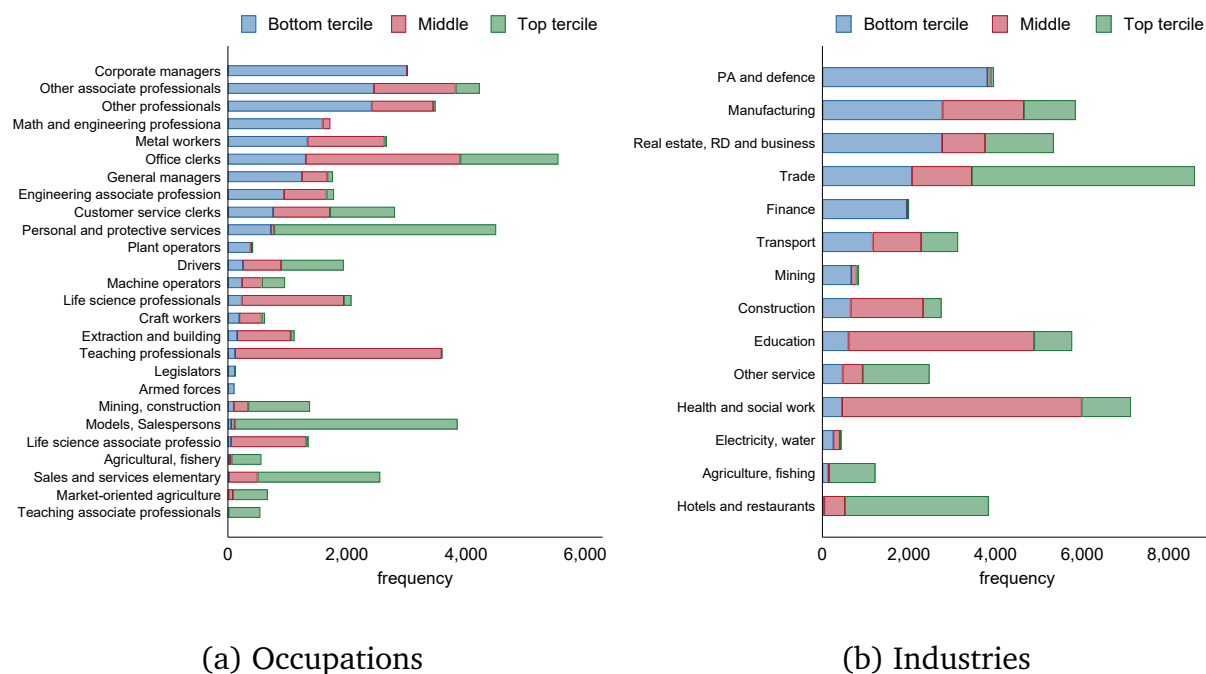
Note: This table replicates Table 2 on the subsample of women who, two years before childbirth, were entitled to employer-provided paid parental leave.

G Terciles of Casual Prevalence

Here we discuss characteristics of the jobs in the terciles of casual prevalence, measured pre-2009. From Table G.1 we notice two main features. First, jobs with lower casual prevalence are associated with higher pay, longer hours, and greater occupational status. Second, while there are differences between jobs in the bottom and in the middle tercile, they are substantially smaller than the differences with jobs in the top tercile. This lends justification to our choice of using women with pre-birth job in the bottom tercile of casual prevalence as our preferred comparison group for treated women in the middle tercile.

The bottom panel of Table G.1 shows that women in jobs in the bottom and in the middle tercile of casual prevalence make similar fertility choices: they both have their first child at 31 on average, while women with jobs in the top tercile of casual prevalence do so at 26; they also have similar fertility patterns. While this similarity is not required by our empirical strategy (our design only requires a triple-difference version of parallel trends, as detailed in Section 4.2.2), it is reassuring on the validity of our strategy.

Figure G.1: Occupations and Industries by Terciles of Casual Prevalence



Note: For each occupation (left panel) and industry (right panel) in our sample, this figure plots the frequency of observations by terciles of casual prevalence. The industries and occupations displayed here are more aggregate than the 2-digit used for the classification. Jobs (occupation-by-industry cells) are weighted by the number of pre-2009 observations, including both women and men.

Table G.1: Job Characteristics by Tercile

	Bottom tercile		Middle tercile		Diff Middle-Bottom		Top tercile		Diff Top-Bottom	
	Mean	SD	Mean	SD	Diff.	SE	Mean	SD	Diff.	SE
<i>All Employees</i>										
Avg hourly wage (2021 dollars)	37.22	(33.50)	31.69	(29.49)	-5.53***	(0.30)	22.57	(24.47)	-14.65***	(0.29)
Avg occupational status	60.34	(19.60)	54.02	(24.04)	-6.32***	(0.21)	30.55	(12.74)	-29.78***	(0.16)
Has bachelor degree or above	0.35	(0.48)	0.32	(0.47)	-0.03***	(0.00)	0.08	(0.27)	-0.27***	(0.00)
Avg weekly working hours	42.75	(12.71)	37.72	(14.46)	-5.04***	(0.13)	30.95	(17.91)	-11.80***	(0.15)
Avg tenure with current employer	7.36	(8.09)	7.22	(8.15)	-0.15	(0.08)	4.92	(7.48)	-2.45***	(0.08)
Share of casual contracts	0.05	(0.21)	0.17	(0.37)	0.12***	(0.00)	0.51	(0.50)	0.46***	(0.00)
Share of regular schedule	0.62	(0.48)	0.48	(0.50)	-0.14***	(0.00)	0.25	(0.43)	-0.38***	(0.00)
Share of on call	0.10	(0.30)	0.10	(0.30)	0.00	(0.00)	0.16	(0.36)	0.06***	(0.00)
Share on flexible start/finish times	0.69	(0.46)	0.48	(0.50)	-0.20***	(0.01)	0.55	(0.50)	-0.14***	(0.01)
Share of women	0.36	(0.48)	0.54	(0.50)	0.18***	(0.00)	0.54	(0.50)	0.18***	(0.00)
Avg Freedom Decision	0.22	(0.33)	0.04	(0.31)	-0.19***	(0.00)	-0.27	(0.39)	-0.50***	(0.00)
Avg Unstructured work	0.22	(0.39)	-0.00	(0.30)	-0.22***	(0.00)	-0.20	(0.41)	-0.41***	(0.00)
Avg Importance of Relationships	0.22	(0.42)	-0.00	(0.33)	-0.23***	(0.00)	-0.23	(0.27)	-0.46***	(0.00)
Avg Time Pressure	-0.01	(0.19)	0.05	(0.21)	0.07***	(0.00)	-0.06	(0.18)	-0.05***	(0.00)
Avg Flex Score	0.17	(0.27)	0.01	(0.21)	-0.16***	(0.00)	-0.21	(0.20)	-0.38***	(0.00)
<i>Women in occupational tercile two years before childbirth</i>										
Avg age at first birth (women)	30.95	(4.69)	30.81	(4.94)	-0.14	(0.52)	26.40	(6.01)	-4.55***	(0.59)
Avg num of children 3 years after first (women)	1.60	(0.54)	1.68	(0.58)	0.07	(0.08)	1.59	(0.55)	-0.02	(0.08)
Avg num of children 5 years after first (women)	1.88	(0.58)	2.01	(0.63)	0.12	(0.08)	1.85	(0.56)	-0.04	(0.08)
Avg num of children 7 years after first (women)	2.00	(0.60)	2.11	(0.74)	0.11	(0.09)	1.95	(0.65)	-0.05	(0.09)

Note: This table shows means and differences in means for various characteristics of jobs, divided into the three terciles of casual prevalence as defined in Section 4.2.2. In the top panel we include all employees (with no distinction of gender) and all observations between 2001 and 2008. In the bottom panel we restrict to women who were in a given occupational tercile two years before the birth of their first child, also between 2001 and 2008. The definition of the main variables is detailed in Appendix A; Goldin's flexibility characteristics (Freedom Decision, Unstructured Work, Importance of Relationship, Time Pressure, and Flex Score) are constructed as follows: for each of these job characteristics, we identify in the HILDA Survey questions that match the O*NET question used by Goldin (2014). The variable "Flex Score" averages them all. Following Goldin (2014), each of these has been coded so that a higher value of the variables denotes a job that has less flexibility, more interpersonal contact, less independence in decision-making, and so on, and thus is "greedier".

H Additional Evidence from the Exposure Design

In this appendix, we provide further evidence on the (weakly positive) career impact of the Fair Work Act, and we rule out a simple explanation of our results based on exposed mothers simply getting closer to their desired number of hours of work. In particular, we do not document statistically significant changes to career trajectories of most exposed mothers, but some specifications point to a gradual improvement. In terms of mechanisms, we show that the increase in hours of work of the most exposed mothers does not bring them closer to their desired number of hours - the earlier cohorts of mothers, with a casual contract, *wanted* to work the fewer hours they did because, with an irregular schedule, every hour worked was more costly (e.g. every hour actually worked corresponding to more than one hour being on call).

Career's Impact of the Fair Work Act Despite the permanent positive effect of the law on maternal labor supply, the Fair Work Act did not have large effects on careers conditional on working, though there is some noisy evidence consistent with a positive effect.

The self-reported probability of having been promoted in the previous 12 months does not change systematically for the most exposed mothers in the seven years following the birth of their first child (column 2 of Table H.1; triple-difference coefficient highlighted in blue). However, there is one noticeable caveat: the event-study in panel (a) of Figure H.1 shows that for the most exposed mothers (middle tercile, the central panel in orange) there is a significant difference in the probability of being promoted in the year following the first childbirth, with later (post-Fair Work Act) cohorts significantly more likely to report no post-childbirth decrease in the probability of promotion. This might be economically meaningful, but since the estimates are noisy, we do not want to over-interpret it.

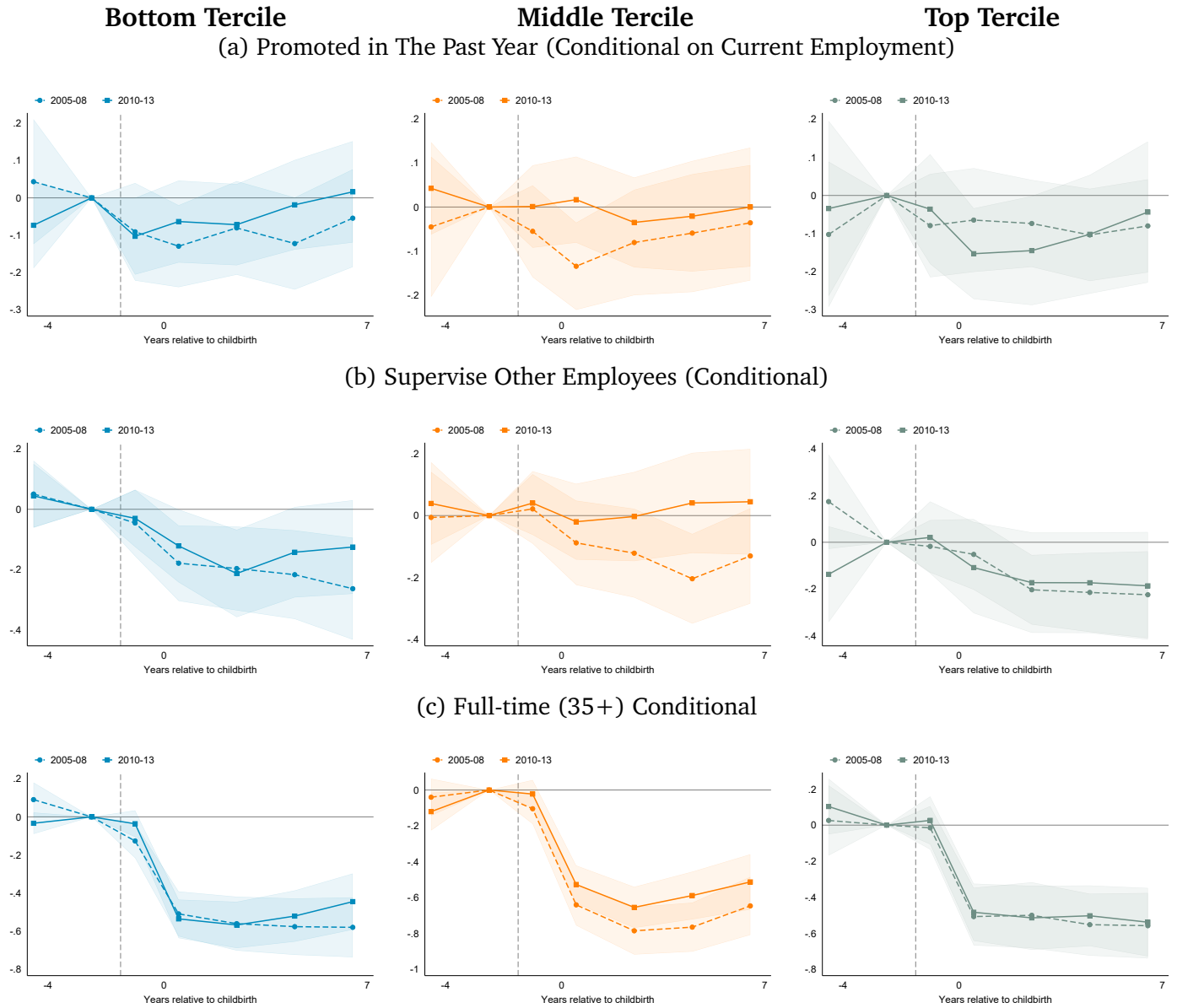
The probability of supervising other employees does not change on average for the most exposed mothers, but the event-study suggests a dynamic positive effect. In column (4) of Table H.1, the point estimates from the triple-difference specification suggest that the child penalty in the probability of having a supervisory role in the seven years following the first childbirth does not change differentially for the more and less exposed mothers. Panel (b) of Figure H.1 though, shows that this average might be hiding a dynamic effect: for the most exposed group (middle tercile, in orange in the middle panel), a progressive shift away from supervisory roles after childbirth evident in the early (pre-Fair Work Act) cohorts is no longer happening for the post-Fair Work Act cohorts. While the evidence is suggestive, the difference is only statistically significant in a single year (five years from first childbirth), and thus we are hesitant to extrapolate a trend from it.

Compliers Would Have Not Counterfactually Worked Full Time We show that mothers who would have counterfactually remained full-time after childbirth are not driven by the Fair Work Act into working less than full-time. After the Fair Work Act, reducing hours is less costly, since it is no longer associated with an irregular schedule: thus, in principle, we could observe women working part-time who would have worked full-time absent the Fair Work Act. Instead, we do not see it happening: there is no decrease in the probability of working full-time; if anything, the point estimates go in the opposite direction. This can be seen in columns (6) and (8) of Table H.1, where we report the estimates of the triple-difference coefficients from Equation (3) (and the coefficients of interest are highlighted in blue), as well as in panel (c) of Figure H.1 (where the most exposed tercile is in orange in the middle).

Mothers Are Not Closer to Their Optimal Hours Here we provide evidence against the hypothesis that permanent contracts, by supposedly offering mothers more hours than the equivalent casual contracts, simply bring them closer to their optimal level of part-time hours. In this piece of analysis, we find that the gap between actual and desired number of hours is not systematically different for mothers in the middle tercile between early and late cohorts; if anything, the point estimate in the triple-difference goes in the opposite direction (column 10 of Table H.1). Similarly, when looking at whether they work too little (the gap between actual and desired hours is negative; column 12), we see that mothers in the middle tercile do not differ systematically between the early and late cohorts, and again the point estimate of the triple-difference points to an (insignificant) *increase* in the probability of working too little relative to desire rather than a decrease, which would be implied by the aforementioned hypothesis. So, we can rule out the explanation that the permanent contracts increase mothers' hours simply by getting them closer to their desired level of hours.

We interpret these results as indicative of a change in mothers' perceived cost of each hour worked part-time. Before the reform, when the main option was a casual contract with irregular hours, mothers' optimal level of hours (i.e., the desired number of hours) was lower, because each hour was more costly with an irregular schedule. After the Fair Work Act, when permanent contracts became more accessible, mothers' desired level of work hours increased, alongside actual worked hours.

Figure H.1: Work Hours and Career Progression of Women around Childbirth, Before and After Reform, by Exposure to the Fair Work Act



Note: This figure shows the child penalty in the probability of having been promoted in the prior year (panel a), in the probability of supervising other employees at work (panel b), and in the probability of working full-time (35 hours per week or more) (panel c) by early and late cohorts and by terciles of casual prevalence. All outcomes are conditional on current employment. See notes to Figures 4 and 5 for details.

Table H.1: Changes in Child Penalty by Exposure to the Fair Work Act - Women (Compliers and Mechanisms)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9) Gap actual- desired weekly working hours	(10) Gap actual- desired weekly working hours	(11) Too little work	(12) Too little work
	Promoted in the past year	Promoted in the past year	Supervise other employees	Supervise other employees	Full-time (35+) cond.	Full-time (35+) cond.	Full-time (35+) uncond.	Full-time (35+) uncond.				
Child Penalty 0-7	-0.0588** (0.0189)	-0.0618+ (0.0325)	-0.154*** (0.0250)	-0.175*** (0.0412)	-0.575*** (0.0247)	-0.532*** (0.0410)	-0.596*** (0.0209)	-0.599*** (0.0347)	-7.775*** (1.705)	-9.905*** (2.170)	0.270*** (0.0546)	0.293*** (0.0704)
* Tercile = 2		0.00337 (0.0453)		0.0558 (0.0589)		-0.122* (0.0567)		-0.0591 (0.0495)		1.664 (3.619)		-0.0214 (0.120)
* Tercile = 3		0.00215 (0.0462)		0.00799 (0.0653)		0.00753 (0.0645)		0.0973+ (0.0517)		13.02* (5.110)		-0.198 (0.175)
Child Penalty 0-7 * Post-2009 First Birth	-0.0102 (0.0218)	0.0159 (0.0383)	0.0575+ (0.0295)	0.0297 (0.0476)	0.0584* (0.0285)	0.0177 (0.0461)	0.0616* (0.0245)	0.00351 (0.0389)	4.643* (2.018)	7.873** (2.613)	-0.127+ (0.0681)	-0.182* (0.0882)
* Tercile = 2		-0.0193 (0.0521)		0.0175 (0.0664)		0.110+ (0.0645)		0.111* (0.0558)		-3.123 (4.138)		0.0874 (0.143)
* Tercile = 3		-0.0604 (0.0552)		0.0781 (0.0827)		0.0268 (0.0763)		0.0818 (0.0625)		-16.52* (6.449)		0.315 (0.225)
Early (pre-2010) cohorts: Mean Y pre-birth	0.160		0.553		0.846		0.767		10.22		0.112	
- tercile 1		0.205		0.621		0.923		0.872		12.26		0.0526
- tercile 2		0.147		0.509		0.838		0.782		12.77		0.0426
- tercile 3		0.113		0.515		0.745		0.618		2.333		0.333
Late (post-2010) cohorts: Mean Y pre-birth	0.167		0.493		0.849		0.780		10.97		0.154	
- tercile 1		0.211		0.552		0.919		0.892		13.75		0.0562
- tercile 2		0.148		0.451		0.852		0.813		11.03		0.139
- tercile 3		0.130		0.479		0.715		0.559		4.286		0.429
Early cohorts: New parents	340		366		365		415		122		122	
- tercile 1		122		128		128		134		51		51
- tercile 2		118		127		127		138		40		40
- tercile 3		100		111		110		143		31		31
Late cohorts: New parents	625		671		670		734		164		164	
- tercile 1		212		223		222		234		57		57
- tercile 2		262		283		283		296		75		75
- tercile 3		151		165		165		204		32		32
Tot observations	20052	20051	23605	23604	23520	23519	31792	31792	3749	3741	3749	3741

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: This table reports the estimates of the child penalty in various aspect of work hours for women and its interaction with treatment status (defined by the tercile of casual prevalence of a woman's modal job in the five years before the birth of her first child). More details in the note to Table 2. The outcome in columns (1) and (2) is a dummy for having been promoted in the previous 12 months (conditional on being employed at the time of answering). The outcome in columns (3) and (4) is a dummy for having a supervisory role at work (conditional on working). The outcome in columns (5) through (8) is a dummy for working full-time (≥ 35 hours per week), both conditionally on working (columns 5 and 6) and unconditionally (columns 7 and 8). The outcome in columns (9) and (10) is the difference between actual weekly hours of work and self-reported desired weekly hours of work. This is only defined on employed individuals and takes the value of zero when actual and desired weekly hours of work coincide. In columns (11) and (12), the outcome variable is a dummy that takes value of one if actual number of hours are less than desired number of hours and zero if they are the same or more.

I Threats to Identification in the Exposure Design

Our job exposure strategy from Section 4.2.2 relies on a triple-difference version of the parallel trend assumption: that the child penalties of mothers in the treated tercile would have evolved, over time, similarly to the ones of the other two groups. We lend support to this assumption in several ways. First, in Section I.1, we show parallel pre-trends in child penalties across terciles. Second, in Section I.2, we rule out selection into childbirth. Third, in Section I.3, we rule out selection into occupations by showing that our results are robust to assigning women the exposure of the job they had before the Fair Work Act. Fourth, in Section I.4, we show that the Fair Work Act did not have an effect on fertility, and thus differential fertility is not a channel for the observed effect on labor supply.

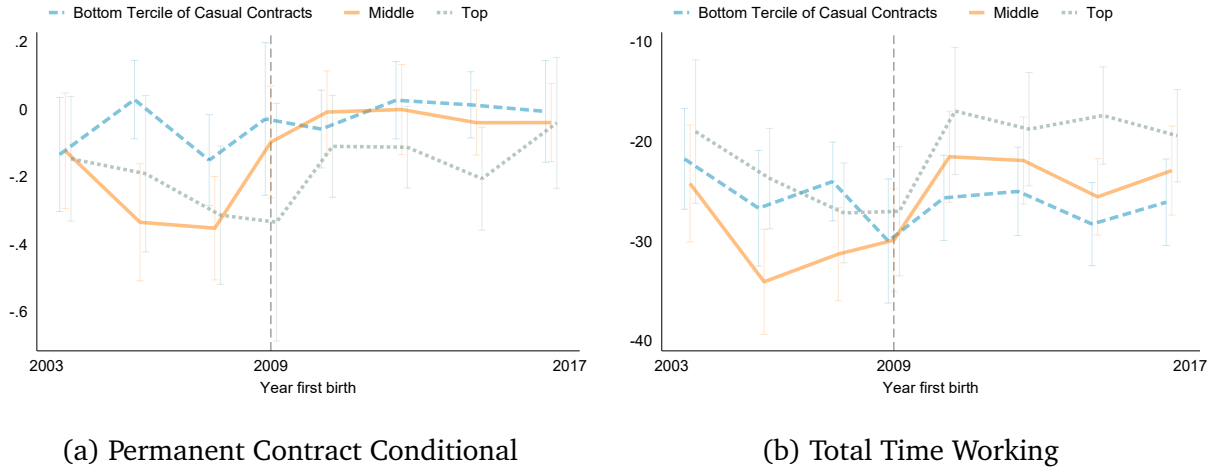
I.1 Parallel Pre-trends

Here we show the evolution over time of the child penalty, separately for the three terciles of casual prevalence to whom mothers belong.

Empirically, we fully interact Equation (5) with tercile-of-casual-prevalence dummies; that is, we replicate Figure E.1 but separately by terciles of casual prevalence. In Figure I.1, the blue dashed line shows the evolution of the child penalty for women with a pre-birth job belonging to the bottom tercile of casual prevalence (main control group), the orange solid line for women with pre-birth jobs in the middle tercile of casual prevalence (treated group), and the dashed grey line for the top tercile. Recall that the bottom and middle tercile are jobs that are closest in characteristics, and thus the orange-versus-blue (bottom versus middle tercile) is the most meaningful comparison.

Child penalties for the bottom and middle tercile of casual prevalence evolve in parallel, except for a break exactly in 2009, which brings them closer. Panel (a) shows the evolution of the child penalty in the probability of having a permanent contract. The blue dashed line shows that this penalty was close to zero throughout the sample period for women in jobs with a very low share of casuals to start with. The orange line shows that instead, for women who gave birth in 2005 or 2007 and had a job in the middle tercile of casual prevalence, there was a sizeable child penalty in this measure (around 30 percentage points), which was relatively stable; there was a sharp decrease in this child penalty around the 2009 cohort, and women with pre-birth jobs in this middle tercile who gave birth from 2010 onwards basically no longer experienced a child penalty in this dimension, exactly like women in the bottom tercile. The child penalties in labor supply (panel b) display analogous patterns.

Figure I.1: Time Evolution of Child Penalty by Exposure to Fair Work Act



Note: This figure shows the evolution of the child penalty in contract type and labor supply for women over the sample period, separately by levels of exposure to the Fair Work Act. The blue dashed lines are estimates for the bottom tercile (lowest casual prevalence), the orange solid lines are estimates for the middle tercile (the most exposed group), the grey dotted lines for the top tercile. Cohorts are biannual (i.e. we pool together women whose birth was in 2003 and in 2004, in 2005 and in 2006, etc; except for 2009, which is kept separate). The vertical segments represent 95% confidence intervals. Women without children and with older children are included as controls, and exposure is assigned based on the modal job they had in the five years before a randomly-drawn fake childbirth year. We cluster standard errors at the individual level.

I.2 Early and Late Cohorts are Similar Pre-Birth in All Terciles

The Fair Work Act did not induce selection into fertility. In Appendix D.2 we discussed how early and late cohorts of will-be mothers are similar on pre-birth observables in the aggregate: this is true also tercile-by-tercile. For example, women whose modal job in the five years before childbirth was in the bottom tercile of casual prevalence on average became mothers at 31 years old both in the early and late cohorts. Furthermore, two years before childbirth, they worked more than 40 hours a week, both in the early and late cohorts, and earned almost 50% of household income, again both in the early and late cohorts. Women whose modal pre-birth job was in the middle tercile of casual prevalence are very similar pre-birth, both between early and late cohorts and compared to women in the bottom tercile. Women in the top tercile are again very similar between early and late cohorts, but they are systematically different from will-be mothers of the other two terciles (for example, they become mothers at a significantly younger age and have significantly lower hourly wages).³⁵

³⁵We don't present the full table here due to space constraints, but it's available upon request (as well as in previous working paper versions of this manuscript, e.g. Ciasullo and Uccioli, 2025, Appendix J.2).

I.3 Robustness to Pre-Fair Work Act Job Choice

Our results are not driven by endogenous job switching in response to the Fair Work Act. The empirical strategy presented in Section 4.2.2 relies on grouping women in terciles of exposure to the Fair Work Act, based on their pre-birth job. Our main specification assigns each woman the modal job she had in the five years prior to the birth of her first child, and women's exposure is measured as the casual prevalence associated with such job. However, if will-be-mothers endogenously switch jobs in response to the Fair Work Act, this might bias our results. Here, we show that this is not the case and that our results are robust to assigning mothers jobs they had chosen before the Fair Work Act.

In columns (1)-(8) of Table I.1, we restrict the sample to women who became mothers in 2007-08 and 2010-11. In columns (1)-(4), we replicate the main columns of Table 2 on this sample, using the same treatment assignment as in the main text. In columns (5)-(8), instead, we base the tercile assignment on the job she held precisely three years before childbirth. This guarantees that all jobs used in the assignment are chosen before 2009, and thus could not be impacted by the Fair Work Act. Comparing column (2) with column (6) and column (4) with column (8) we see that the two specifications yield basically identical results, indicating that endogenous job switching is not driving our main results.

I.4 No Effects on Fertility or Via Fertility

The Fair Work Act did not induce differential fertility, and thus the estimated effects are not mediated by exposed mothers having more or fewer children in response to the law. Column (10) of Table I.1 shows that mothers exposed to the Fair Work Act do not have a different number of children.³⁶ Consistently, our results replicate when conditioning on a completed fertility of one, albeit more noisily, since only about 20% of mothers in our sample satisfy this restriction (results not shown, available upon request).

This evidence should mitigate concerns that differential birth timing across skill groups, as discussed in Adams et al. (2024), would drive the estimated results. Since we find no effect on fertility, differential fertility cannot be the channel through which the Fair Work Act operates. In addition, the differences that might generate bias in the estimation of the child penalty are held constant in our preferred comparison: mothers with pre-birth jobs in the bottom and in the middle tercile of casual prevalence have similar education, similar age at first birth, and similar fertility patterns (as described in Appendix G).

³⁶A more complete version of this table, showing no differential probability of birth or pregnancy and of completed fertility three, five, and seven years after the first child, is available in a longer working paper version (Ciasullo and Uccioli, 2025, Appendix J.4).

Table I.1: Robustness to Only Using Jobs Chosen Before Fair Work Act + Fertility

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Main Treatment Definition				Alternative Treatment Definition				Fertility	
	Permanent contract (uncond)	Permanent contract (uncond)	Total time work	Total time work	Permanent contract (uncond)	Permanent contract (uncond)	Total time work	Total time work	Total children ever had	Total children ever had
Child Penalty 0-7	-0.388*** (0.0371)	-0.323*** (0.0662)	-28.42*** (1.301)	-25.28*** (2.034)	-0.388*** (0.0371)	-0.333*** (0.0694)	-28.44*** (1.301)	-25.30*** (2.072)	1.177*** (0.0331)	1.222*** (0.0473)
* Tercile = 2		-0.162+ (0.0863)		-8.245** (2.915)		-0.152+ (0.0877)		-8.640** (2.927)		-0.0506 (0.0730)
* Tercile = 3		-0.00468 (0.0982)		-0.277 (3.114)		0.00937 (0.0971)		-0.444 (3.102)		-0.105 (0.0796)
Child Penalty 0-7 * Post-2009 First Birth	0.186*** (0.0531)	0.0540 (0.0852)	6.284*** (1.854)	0.599 (2.772)	0.186*** (0.0531)	0.0218 (0.0869)	6.315*** (1.854)	0.188 (2.699)	-0.176*** (0.0259)	-0.175*** (0.0416)
* Tercile = 2		0.283* (0.118)		10.39** (3.894)		0.351** (0.118)		10.70** (3.884)		-0.0604 (0.0610)
* Tercile = 3		0.0992 (0.138)		7.954 (5.330)		0.117 (0.140)		7.895 (5.223)		0.0677 (0.0650)
Early (pre-2010) cohorts: Mean Y pre-birth	0.737		42.70		0.737		42.70		0	
- tercile 1		0.861		44.31		0.833		43.89		0
- tercile 2		0.787		43.50		0.806		43.25		0
- tercile 3		0.505		39.39		0.555		40.81		0
Late (post-2010) cohorts: Mean Y pre-birth	0.747		41.93		0.747		41.93		0	
- tercile 1		0.860		44.40		0.836		44.52		0
- tercile 2		0.718		40.75		0.752		41.16		0
- tercile 3		0.597		39.70		0.575		38.14		0
Early cohorts: New parents	112		108		112		108		415	
- tercile 1		34		33		33		31		134
- tercile 2		42		41		40		38		138
- tercile 3		36		34		39		39		143
Late cohorts: New parents	116		108		116		108		734	
- tercile 1		41		39		41		40		234
- tercile 2		46		46		42		42		296
- tercile 3		29		23		33		26		204
Tot observations	17437	17434	14974	14971	17445	17444	14984	14981	31792	31792

Standard errors in parentheses

+ $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: In columns (1)-(8), this table replicates the primary columns of Table 2 on a smaller sample (focusing on cohorts of women who had their first child in 2007-08 and 2010-11) and using alternative definitions of exposure. Columns (1)-(4) use the definition of exposure we use in the main specification (each woman is assigned the modal job she had in the five years prior to the birth of her first child and the casual prevalence associated with it), while columns (5)-(8) assign each woman the job she had three years before the birth of her first child. This ensures that even in the post-Fair Work Act cohorts, women are assigned a job they held before the Fair Work Act.

In columns (9) and (10) we use the main sample of women (same sample and specification as in Table 2, see relative table note for details) but using as outcome the total number of children the respondent ever had (note that this is time-varying within individuals).